

Code-Aware Joint Estimation and Detection for Doppler-Induced ICI in Underwater Acoustic OFDM: Measured-Channel Validation

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Abstract—Residual Doppler makes underwater acoustic OFDM depart from the diagonal subcarrier model: intercarrier interference appears, and a pilot-only equalizer can land just outside the decoder’s basin of attraction. We present JUNA (Joint Unified Nonideality-Aware Algorithm), a code-aware receiver family that places equalization and LDPC decoding in one coupled objective. The measured gradient receiver, JUNA-Wz, freezes the explicit ICI tensor and takes confidence-weighted descent steps over the partial-FFT combiner and soft bit latents, with LDPC belief-propagation posteriors providing parity structure and per-symbol confidence. Unlike path-based or banded-ICI channel-estimation receivers, JUNA does not deploy a full frequency-domain ICI-matrix estimator; that model motivates the update and is exercised separately as a diagnostic. A lightweight variant, JUNA-lite, approximates the same code-aware feedback loop without explicit gradients—it feeds LDPC posterior means back into the partial-FFT RLS combiner as soft data anchors and reuses the front end—recovering most of the gain at a fraction of the cost, which makes a large measured-channel study tractable. At 20 dB, JUNA-lite+FEC improves packet success over Partial FFT+FEC in hard-but-decodable measured channels: from 0.559 to 0.922 payload-exact PSR on the controlled red channel and from 18/190 to 135/190 accepted packets on the hardest native capture, while tying on ceiling channels and failing with the baselines on floor channels. In stacked head-to-head mirrors, JUNA-Wz is at least as good as JUNA-lite in 44/48 capture-SNR cells, has 18 disjoint Wilson-interval wins and no disjoint losses, and has four small non-significant regressions on near-floor NA cells. A twelve-capture, three-site tuning study shows that FFT size is usually the first high-impact retuning knob, after which conservative modulation and code-rate choices keep the receiver inside the decodable regime.

Index Terms—Underwater acoustic communication, OFDM, intercarrier interference, Doppler, joint detection and decoding, LDPC, partial-FFT, recursive least squares, replay channel.

I. INTRODUCTION

UNDERWATER acoustic (UWA) OFDM is attractive because it turns a long multipath channel into many narrowband subchannels when timing, Doppler correction, and cyclic-prefix assumptions hold [1]. The same structure becomes fragile when the channel changes over the useful OFDM interval. Even after packet acquisition and coarse Doppler correction, residual time variation breaks subcarrier orthogonality, spreads energy into neighboring bins [2], and leaves the receiver with a coupled-carrier problem rather than a diagonal one.

Partial-FFT demodulation is a serious conventional defense against this breakdown [3]. Instead of representing each subcarrier by a single FFT coefficient, the receiver slices the OFDM symbol in time, computes several partial-FFT observations per carrier, and learns a local linear combiner—here a bandwise combiner trained by recursive least squares (RLS) on known pilots. This baseline is far stronger than one-tap OFDM; in our replay workbench it is the strongest non-decoder-feedback receiver. Its limitation is structural: it trains the equalizer from pilots and only then asks the LDPC decoder whether the resulting soft symbols make sense. When pilot support is sparse or the time variation is strong, the pilot-only combiner can land just outside the decoding basin, even though the decoder posterior often carries useful global information about the codeword.

JUNA uses that information before giving up. The formulation treats the partial-FFT combiner, transmitted data symbols, and code constraints as parts of one regularized problem. The measured gradient receiver, JUNA-Wz, uses the partial-FFT observation vectors and a frozen ICI model, then jointly nudges the combiner and the soft bit latents toward a configuration consistent with the pilots and—through the LDPC belief-propagation (BP) posterior—the code. This is intentionally close in spirit to turbo equalization [4]–[6], but architecturally narrower (Section I-A): JUNA does not replace the front end with a generic soft-in/soft-out equalizer; it keeps the partial-FFT observation vectors and feeds decoder-weighted information back into that front end. A lightweight variant, JUNA-lite, approximates the same loop without forming any gradient: it maps the posterior LLRs to posterior-mean constellation points and re-estimates the same partial-FFT RLS combiner with those means as cap-limited soft training anchors. The key economy—shared by both—is that a decoded data symbol with a useful posterior can behave like a *soft pilot*: the LDPC parity graph supplies structure beyond the physical pilot tones and can share the equalizer-training burden. Because JUNA-lite is cheap enough to run across every measured capture, it carries the large-scale study, while JUNA-Wz is reported on head-to-head mirrors and JUNA-WCz remains a diagnostic.

Contributions.

- A coupled estimation/decoding objective (Section IV) in which pilots, partial-FFT observations, the combiner, and the code constraints regularize one another.
- Two bounded receivers derived from it (Section V): JUNA-Wz, which descends the reduced $(\mathbf{W}, \mathbf{z})$ objec-

tive with confidence-weighted updates and optional BP projection, and a lightweight JUNA-lite that approximates the same loop as a posterior-anchor RLS refit (partial-FFT RLS seed $\rightarrow$ BP posterior LLRs $\rightarrow$ posterior-mean soft data anchors $\rightarrow$ combiner refit $\rightarrow$ candidate selection), with explicit anchor-count and iteration limits. The full $(\mathbf{W}, \mathbf{C}, \mathbf{z})$ solver, JUNA-WCz, is reported as a diagnostic rather than as the deployed sweep receiver.

- A complexity accounting (Section VI) that separates the online receiver cost from the offline configuration search and quantifies the gradient-versus-lite trade.
- An empirical regime characterization: JUNA helps in the hard-but-decodable regime, ties on ceiling channels, and fails with the baselines on floor channels, with JUNA-lite carrying the at-scale sweeps and JUNA-Wz tested on head-to-head mirrors.
- Validation on measured replay captures across three deployment sites (Section IX) and actionable cross-channel tuning guidance.

A. Related Work

UWA-OFDM receiver design has largely followed an estimate-then-decode pattern. One line of work estimates a physical or sparse channel model and derives the resulting non-diagonal, often locally banded, ICI pattern explicitly [2]; another tracks time-varying channels across blocks by exploiting channel coherence [7]. These papers explain the physical source of the problem studied here, but they are not the same receiver architecture: they seek explicit channel or ICI estimates, whereas JUNA-Wz descends a code-aware reduced objective over an empirical partial-FFT combiner and the data symbols using decoder-derived confidences (and its JUNA-lite variant retrains that combiner from decoder-derived soft targets).

Other approaches attack Doppler before this receiver loop by changing the transmit waveform or the estimation basis. Doppler-resilient orthogonal signal-division multiplexing (D-OSDM) designs the data/pilot vector layout so that, under a basis-expansion model of the doubly spread channel, the block structure is preserved after propagation and Doppler-induced coupling can be inverted at the receiver [8]. A second group changes the observation or estimation basis rather than the modulation: frequency-domain oversampling of zero-padded OFDM sharpens channel/ICI estimation and reduces sensitivity to Doppler leakage without altering the transmit waveform [9], and wavelet/filter-bank representations have been used to refine channel estimation under Doppler spread [10]. These methods share JUNA's motivation—all respond to the same Doppler-created off-diagonal coupling—but move the remedy into the transmit waveform or an alternative channel-estimation basis. JUNA instead keeps the measured CP-OFDM/Partial-FFT front end and asks whether decoder posterior structure can supply additional soft anchors once residual coupling has already reached the receiver.

Partial-FFT demodulation [3] attacks intra-symbol time variation directly and is the front end we adopt. It replaces one FFT coefficient per carrier with several time-windowed

partial-FFT observations and learns a linear combiner from pilots. JUNA keeps this observation-vector representation; the new step is to let the decoder posterior either weight a reduced gradient update of the combiner and the soft data symbols (JUNA-Wz) or provide soft targets for a bounded anchored RLS refit (JUNA-lite). Pilot-aided ICI mitigation outside UWA similarly motivates the diagonal-to-coupled transition [11], but does not provide this decoder-aided partial-FFT retraining loop.

The closest algorithmic relatives are iterative, turbo-principle ICI receivers for time-varying UWA channels [6], [12]. They exchange extrinsic LLRs between an ICI-cancelling equalizer and the decoder. JUNA makes a narrower exchange: decoder confidences *weight* the JUNA-Wz update of the partial-FFT combiner and soft data symbols—and, in the JUNA-lite variant, convert decoder *posterior means* into soft data pilots that retrain the RLS combiner—under anchor/iteration caps, optional confidence thresholding, and decoder-grounded candidate selection. A broader family of iterative UWA-OFDM receivers couples detection with external interference cancellation [13], physical-layer network coding at a relay [14], or asynchronous multiuser reception [15]; these solve different interference or network problems. To our knowledge no prior UWA-OFDM receiver feeds decoder posteriors back into the partial-FFT combiner as soft anchors in this way.

The experiments below therefore make a deliberately narrower comparison: they isolate the incremental value of posterior-anchor retraining over the one-tap and partial-FFT receiver variants available in the replay workbench. We do not claim an empirical win over full turbo ICI cancellers or path-explicit ICI estimators; implementing those receivers on the same measured captures is an important but separate comparison. Fig. 1 summarizes the architectural distinction.

II. OFDM UNDER RESIDUAL DOPPLER

A. Diagonal CP-OFDM

Consider one OFDM symbol with N subcarriers, cyclic prefix (CP) length N_{cp} , pilot set $\mathcal{P}$, data set $\mathcal{D}$, and unitary DFT matrix $\mathbf{F}$. Let $\mathbf{X} \in \mathbb{C}^N$ be the transmitted frequency-domain symbol and $\mathbf{x} = \mathbf{F}^H \mathbf{X}$ its time-domain block. If the channel is L -tap, time invariant over the block, and $L - 1 \leq N_{cp}$, CP removal yields a circulant model and, after the FFT,

$$\mathbf{Y} = \mathbf{F} \mathbf{H}_{\text{cir}} \mathbf{F}^H \mathbf{X} + \boldsymbol{\nu} = \text{diag}(\mathbf{H}) \mathbf{X} + \boldsymbol{\nu}, \quad (1)$$

so each carrier obeys the one-tap model $Y_k = H_k X_k + \nu_k$, and pilot-linear equalization estimates H_k on pilots and interpolates.

B. Residual Doppler and Non-Diagonal Coupling

Under Doppler the received packet is time-scaled, $r(t) = s((1+a)t)$, with a the residual scale after coarse correction. This single-scale form is a local approximation; measured channels also exhibit nonuniform Doppler and time-varying multipath [1], so the implementation does not assume one

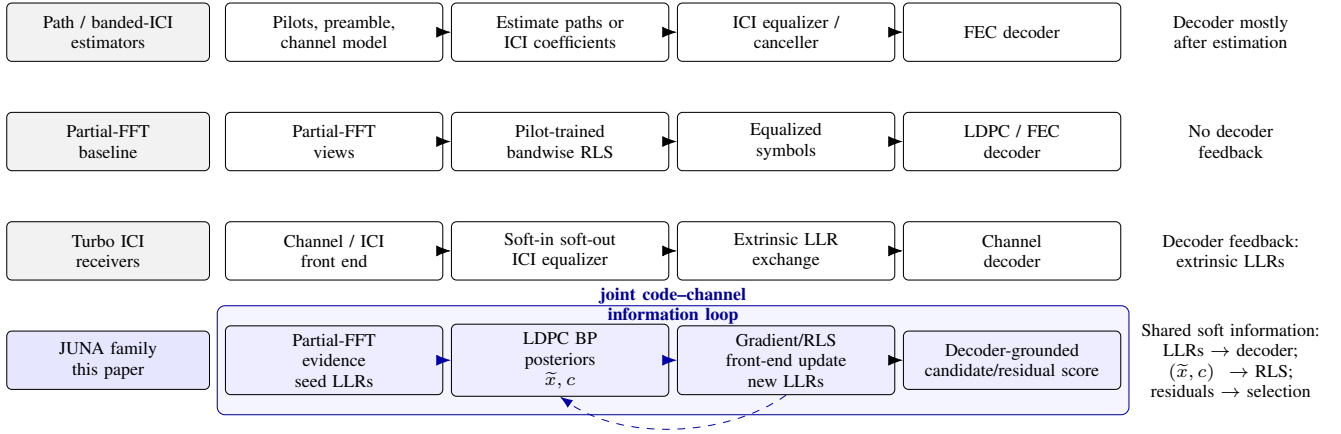


Fig. 1. Receiver information-flow comparison. Path-explicit and banded-ICI methods estimate channel/ICI coefficients before decoding; the Partial-FFT baseline trains a partial-FFT combiner only from pilots; turbo ICI receivers exchange extrinsic LLRs with a soft equalizer; the JUNA family keeps the Partial-FFT front end and closes the shaded joint code-channel loop by sharing soft information. Partial-FFT observation-vector evidence produces seed LLRs, LDPC BP returns posterior means and confidences, JUNA-Wz uses them in a reduced gradient update while JUNA-lite uses them as RLS soft anchors, and updated LLRs/residual evidence return to the decoder without deploying a full ICI-matrix estimator.

global scale. The origin of ICI is already visible in a stripped-down single-path picture. A received FFT bin is an inner product against one receiver-side Fourier tone; for transmitted carrier ℓ and received bin k , a residual scale leaves a term of the form

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{j2\pi((1+a)\ell-k)n/N}. \quad (2)$$

When $a = 0$, the Fourier sums close over the FFT window and the expression vanishes for $k \neq \ell$. When $a \neq 0$, the tones are no longer exactly on the receiver's frequency grid, the cancellation is incomplete, and neighboring columns leak into row k of the frequency-domain channel matrix. Fig. 2 is the same statement in matrix form: clean CP-OFDM is diagonal, while residual Doppler/time variation turns the diagonal operator into a locally coupled one. For pure time scaling the leakage magnitude is a Dirichlet kernel in $(1+a)\ell - k$, so its peak lies near the shifted index $k \approx (1+a)\ell$. The banded near-diagonal picture below therefore assumes that coarse Doppler correction has made the residual scale small over the active band; (2) also suppresses the passband carrier shift and keeps only the subcarrier-index term.

Let $\mathbf{G}$ be the post-acquisition, CP-removed linear operator over one block. Then $\mathbf{Y} = \mathbf{F}\mathbf{G}\mathbf{F}^H\mathbf{X} + \boldsymbol{\nu} \triangleq \mathbf{H}_{\text{fd}}\mathbf{X} + \boldsymbol{\nu}$, and if $\mathbf{G}$ is not circulant, $\mathbf{H}_{\text{fd}}$ is not diagonal:

$$Y_k = H_{k,k}X_k + \sum_{\ell \in \mathcal{B}_K(k) \setminus \{k\}} H_{k,\ell}X_\ell + \epsilon_k^{(K)} + \nu_k, \quad (3)$$

with ICI neighborhood $\mathcal{B}_K(k) = \{\ell : |\ell - k| \leq K\}$. After averaging over independent zero-mean coded data symbols, the truncation residual has energy $\mathbb{E}|\epsilon_k^{(K)}|^2 = \sum_{\ell \notin \mathcal{B}_K(k)} |H_{k,\ell}|^2 \mathbb{E}|X_\ell|^2$; deterministic pilots introduce cross-terms unless averaged over pilot locations or payload realizations. For local residual Doppler the off-diagonal energy concentrates near the main diagonal, so the banded form is accurate at modest K . Equation (3) is precisely why a one-tap receiver fails: it fits a diagonal

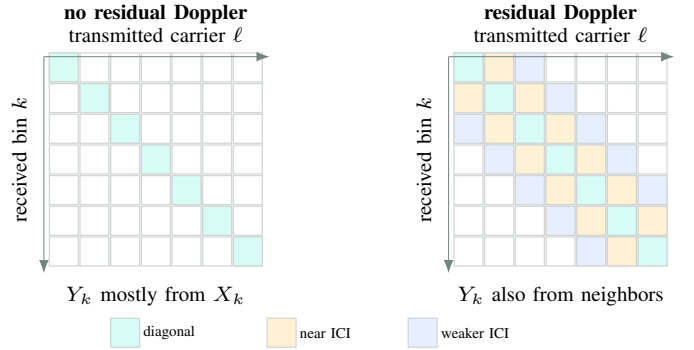


Fig. 2. Origin of the ICI matrix. With an adequate CP and no residual Doppler, the DFT diagonalizes the block channel, so each row of $\mathbf{H}_{\text{fd}}$ has energy mainly on its matching column. Residual Doppler/time variation breaks that Fourier cancellation; row k receives energy from nearby columns, giving the off-diagonal ICI terms in (3).

model while the physical channel leaves a locally coupled operator. The off-diagonal energy—the channel's *dynamic non-diagonality*—is the physical diagnostic motivating code-aware refinement, although this paper does not estimate it as a per-capture predictor. In this paper the banded form is a diagnostic model, not the implemented estimator. Banded-ICI and sparse-channel receivers attempt to recover the coupling coefficients $H_{k,\ell}$ or the path parameters that generate them; the measured-channel JUNA receiver instead observes partial-FFT observation vectors and learns a local scalar combiner from pilots and decoder-derived soft anchors.

The same notation also covers finite-CP stress. If the CP is shorter than the effective delay spread, or if the channel varies across the useful interval, the post-CP operator is not circulant. For OFDM symbol s one can write

$$\mathbf{Y}_s = \mathbf{H}_0\mathbf{X}_s + \mathbf{J}_-\mathbf{X}_{s-1} + \mathbf{J}_+\mathbf{X}_{s+1} + \boldsymbol{\nu}_s, \quad (4)$$

where $\mathbf{H}_0$ is generally non-diagonal and $\mathbf{J}_\pm$ collect inter-symbol leakage from neighboring OFDM symbols. Thus Doppler ICI and CP shortfall both appear to the receiver as

coupled-carrier/coupled-symbol interference. JUNA does not eliminate the need for a CP, but if this residual coupling remains inside the partial-FFT combiner's effective model class, code constraints can help the front end suppress it instead of relying only on pilots.

III. BACKGROUND: PARTIAL-FFT EQUALIZATION

A. Partial-FFT Observation Vectors

Partial-FFT reception retains information about intra-symbol time variation [3]. With segment windows $g_p[n]$, $p = 1, \dots, N_{\text{view}}$, the p -th observation for carrier k is $y_{p,k} = \frac{1}{\sqrt{N}} \sum_n g_p[n] r[n] e^{-j2\pi kn/N}$, stacked into $\mathbf{y}_k \in \mathbb{C}^{N_{\text{view}}}$. If the windows form a partition of unity then $\sum_p y_{p,k} = Y_k$: the partial FFT invents no energy, it decomposes the full FFT statistic into temporal views whose relative phases and amplitudes carry the time-variation information.

B. Pilot-Trained RLS Combiner

The Partial FFT+FEC baseline forms a scalar soft symbol with a bandwise linear combiner $\hat{x}_{k,s} = \mathbf{w}_b^H \mathbf{y}_{k,s}$, $k \in b$. In the reported implementation, one combiner per frequency band is trained over the current packet/candidate using pilot anchors $(k, s) \in \mathcal{P}_b$ against the known $x_{k,s}$:

$$\hat{\mathbf{w}}_b = \arg \min_{\mathbf{w}} \sum_{(k,s) \in \mathcal{P}_b} \lambda^{\tau_{k,s}} |x_{k,s} - \mathbf{w}^H \mathbf{y}_{k,s}|^2 + \delta \|\mathbf{w}\|_2^2, \quad (5)$$

with forgetting factor λ , sample age $\tau_{k,s}$ ($\tau = 0$ for the most recent RLS update), and ridge δ . The recursion is initialized as $\mathbf{w}_{b,0} = \mathbf{0}$ and $\mathbf{R}_{b,0} = \delta^{-1} \mathbf{I}$, then updated with gain $\mathbf{g}_i = \mathbf{R}_{i-1} \mathbf{y}_i / (\lambda + \mathbf{y}_i^H \mathbf{R}_{i-1} \mathbf{y}_i)$, $\mathbf{w}_i = \mathbf{w}_{i-1} + \mathbf{g}_i e_i^*$, and $\mathbf{R}_i = \lambda^{-1} (\mathbf{R}_{i-1} - \mathbf{g}_i \mathbf{y}_i^H \mathbf{R}_{i-1})$. Partial FFT+FEC is thus a strong baseline—time-window diversity plus adaptive pilot-trained combining—but it remains a pilot-only equalizer.

IV. JUNA: A COUPLED OBJECTIVE

The pilot-trained combiner of Section III fixes its weights from pilots before decoding, so the LDPC code acts only after the soft symbols have been formed. JUNA removes that ordering at the level of the objective: it replaces the pilot-stacked desired response with a packet-level forward model and adds a differentiable parity penalty, so that pilots, partial-FFT observations, the combiner, and the code constraints regularize one another within a single problem. We state that ideal objective here; Section V describes the bounded alternating procedure actually run on measured channels, which we do not claim to be an exact minimizer of it.

A. Packet Layout and Tanner-Graph Example

Fig. 3 reproduces the workbench packet/Tanner layout for a concrete configuration, n1024_cp256_sym1_p3_r0p25_dc4_sig1_ip2. The left panel separates the physical OFDM layer from the logical FEC layer. Outer pilots are frequency-domain comb pilots on known subcarriers. Inner pilots are known positions in the serialized message row before LDPC encoding; they are drawn as violet anchors in the code stream, then transmitted

on ordinary data tones after encoding and modulation. The right panel shows the corresponding Tanner-graph slice for the systematic implementation $\mathbf{H}_c = [\mathbf{H}_m \mathbf{I}_m]$. The suffix dc4 is a configuration sparsity knob $d_{\text{cfg}} = 4$; at this rate the constructor maps it to $d_v = 6$ message-column edges in the Tanner graph.

B. Bit Latents, Soft QPSK Symbols, and a Parity Surrogate

An LDPC code is fixed by a sparse parity-check matrix $\mathbf{H}_c \in \mathbb{F}_2^{m \times n_b}$ with n_b coded bits and m checks [16]; an encoder maps $\mathbf{u}$ to a codeword $\mathbf{b}$ with $\mathbf{H}_c \mathbf{b}^T = \mathbf{0}$ over $\mathbb{F}_2$. Belief propagation (BP) on the induced Tanner graph remains the hard-decision decoder [17]. The implementation uses a systematic construction $\mathbf{H}_c = [\mathbf{H}_m \mathbf{I}_m]$: k message bits are mapped to $n_b = k + m$ coded bits, with rate k/n_b set by the modem configuration and sparse message-bit degree chosen from the configured sparsity knob d_{cfg} . For example, the workbench suffix dc4 means $d_{\text{cfg}} = 4$; under the rate-0.25 rule used for Fig. 3, this becomes actual message-column degree $d_v = 6$. Some message positions can be fixed as *inner pilots*; these are known bits inside the LDPC message, distinct from outer OFDM pilot tones, and their LLRs are clamped during BP. The payload BER and PSR reported below exclude these inner-pilot message positions. Thus the LDPC design rate is k/n_b , while the net unknown payload rate is lower when inner-pilot message positions are reserved as known anchors.

The identity parity tail makes $\mathbf{H}_c$ full row rank by construction and gives a systematic encoder: for message vector $\mathbf{u}$, the parity suffix is $\mathbf{v} = \mathbf{H}_m \mathbf{u} \pmod{2}$, so $[\mathbf{u}, \mathbf{v}]$ satisfies $\mathbf{H}_c \mathbf{b}^T = \mathbf{0}$. This code is intentionally a simple, deterministic workbench sparse parity code rather than an optimized acoustic LDPC design. The random sparse message part is generated from the fixed packet/configuration seed, and all receiver variants in a comparison use the identical code instance; therefore the measured gains should be interpreted as receiver-level gains under this fixed sparse code.

Because the code constraints are binary while the subcarriers carry QPSK, the bit and symbol variables are kept separate. Let $\xi_i = \tanh(z_i/2) \in (-1, 1)$ be the relaxed bipolar variable for coded bit b_i , so hard bits satisfy $\xi_i = 1 - 2b_i$. For Gray-labeled QPSK, data carrier k carries bits $i_1(k), i_2(k)$ and its relaxed symbol is

$$s_k(\mathbf{z}) = \frac{1}{\sqrt{2}} (\xi_{i_1(k)}(\mathbf{z}) + j \xi_{i_2(k)}(\mathbf{z})), \quad (6)$$

while pilots use known $s_p = P_p$. With check neighborhoods $\{\mathcal{N}_j\}$, the differentiable parity surrogate is

$$\mathcal{R}_{\text{par}}(\mathbf{z}) = \sum_{j=1}^m \left(1 - \prod_{i \in \mathcal{N}_j} \xi_i(\mathbf{z}) \right)^2, \quad (7)$$

which equals 0 exactly when every check is satisfied for hard bipolar bits, so $\mathcal{R}_{\text{par}}$ is a smooth code regularizer that preserves the binary/complex distinction. This surrogate is used as a differentiable consistency penalty in the objective lens and gradient diagnostics; the implemented receiver relies on BP for the actual code update because direct optimization of high-degree parity products can be poorly conditioned and can have weak gradients.

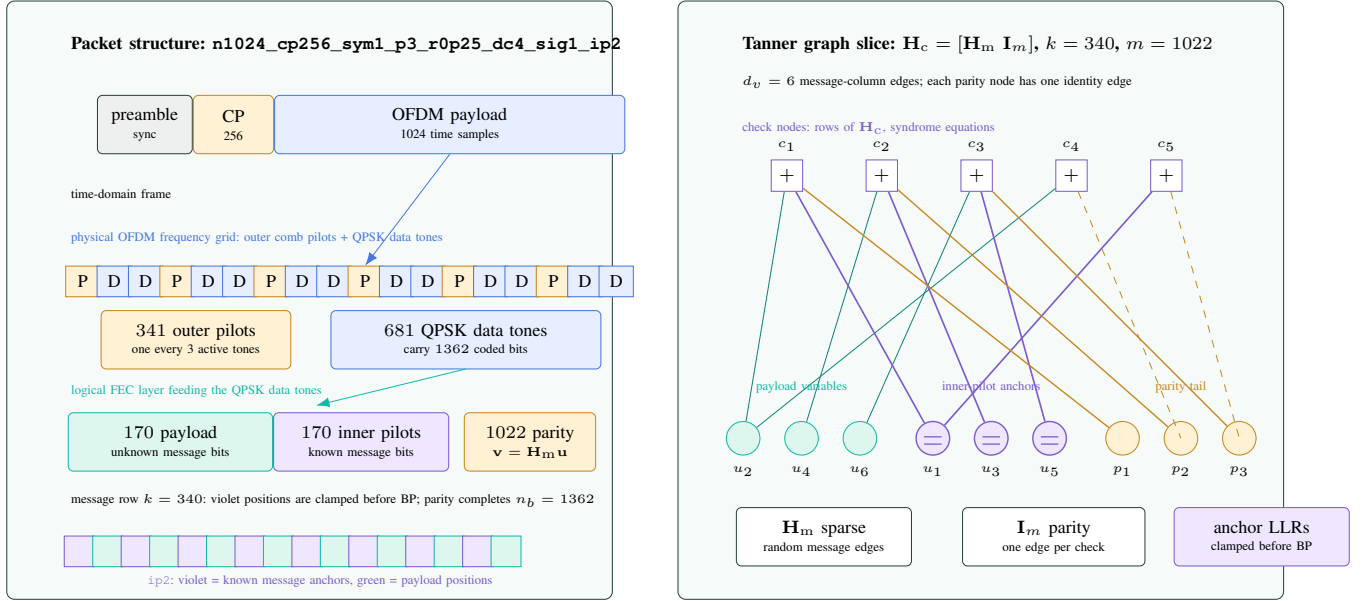


Fig. 3. Workbench-style packet-structure and Tanner-graph example for n1024_cp256_sym1_p3_r0p25_dc4_sig1_ip2. The physical OFDM layer has 1022 active tones per symbol, split into 341 outer frequency-domain pilots and 681 QPSK data tones. The logical FEC layer maps $k = 340$ systematic message bits and $m = 1022$ parity bits to $n_b = 1362$ coded bits. Inner pilots are known positions inside the message row (170 anchors for ip2); they are not extra OFDM pilot tones. In the Tanner graph, violet anchor variables enter BP with clamped LLRs, green payload variables are unknown, and gold parity variables form the identity tail of $\mathbf{H}_c = [\mathbf{H}_m \mathbf{I}_m]$. The illustrated layout is QPSK-specific: for BPSK, the same mapping is used but each data tone carries one coded bit, so $n_b = N_d$ rather than $2N_d$.

This parity structure is the mathematical reason code information can provide training equations beyond the physical pilots. With pilots alone, the combiner in band b is identified only from $\mathcal{P}_b$. After BP, a data carrier with posterior mean $\tilde{x}_{d,s} = \mathbb{E}[X_{d,s} | \mathbf{Y}]$ and confidence $c_{d,s}$ can contribute an additional soft equation. A weighted batch view of the update is

$$\begin{aligned} \hat{\mathbf{w}}_b = \arg \min_{\mathbf{w}} & \sum_{(p,s) \in \mathcal{P}_b} \lambda_{\text{RLS}}^{\tau_{p,s}} |P_{p,s} - \mathbf{w}^H \mathbf{y}_{p,s}|^2 \\ & + \rho \sum_{(d,s) \in \mathcal{D}_b^+} \omega_{d,s} \lambda_{\text{RLS}}^{\tau_{d,s}} |\tilde{x}_{d,s} - \mathbf{w}^H \mathbf{y}_{d,s}|^2 \quad (8) \\ & + \delta \|\mathbf{w}\|_2^2, \end{aligned}$$

where $\mathcal{D}_b^+$ is the selected data-anchor set, $\omega_{d,s}$ is an optional anchor weight, τ is RLS update age ($\tau = 0$ for the most recent sample), and λ_{RLS} is the forgetting factor. The measured receiver sets $\omega_{d,s} = 1$ for selected anchors; confidence $c_{d,s}$ gates and ranks candidate anchors when a threshold or cap is active, while uncertainty is already expressed by the shrunk posterior mean $\tilde{x}_{d,s}$. Compared with (5), the normal matrix gains selected data-anchor terms $\sum_{(d,s) \in \mathcal{D}_b^+} \omega_{d,s} \mathbf{y}_{d,s} \mathbf{y}_{d,s}^H$. When the decoder posterior is informative, those terms improve conditioning just as additional pilots would; when it is uninformative or biased, retained physical pilots, posterior means rather than hard slices, anchor caps, optional confidence thresholding/ranking, and candidate selection limit the damage. In the profiled QPSK benchmark, protection comes mainly from posterior means, retained pilots, the fixed anchor cap, candidate selection, and early stopping: $c_{\min} = 0$ disables the confidence threshold itself, so confidence only affects cap-based ranking when the cap binds.

C. The Coupled Objective

Let $\mathbf{Y}_k \in \mathbb{C}^M$ collect the M partial-FFT observations at carrier k , $\mathbf{w}_k$ the combiner forming $\hat{s}_k = \mathbf{w}_k^H \mathbf{Y}_k$, and $\mathbf{c}_{k,q} \in \mathbb{C}^M$ the partial-FFT-domain response from carrier q to k . With banded support $\mathcal{B}_K(k)$, the forward model is $\mathbf{Y}_k = \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) + \mathbf{n}_k$, and the ideal joint surrogate is

$$\begin{aligned} \mathcal{L}(\mathbf{C}, \mathbf{W}, \mathbf{z}) = & \lambda_d \sum_{k \in \mathcal{K}} \left\| \mathbf{Y}_k - \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) \right\|_{\Sigma_k^{-1}}^2 \\ & + \eta \sum_{p \in \mathcal{P}} |\mathbf{w}_p^H \mathbf{Y}_p - P_p|^2 \\ & + \rho \sum_{k \in \mathcal{D}} |\mathbf{w}_k^H \mathbf{Y}_k - s_k(\mathbf{z})|^2 \quad (9) \\ & + \gamma_C \|\mathbf{C}\|_F^2 + \gamma_W \sum_k \|\mathbf{w}_k\|_2^2 \\ & + \gamma_S \sum_k \|\mathbf{w}_{k+1} - \mathbf{w}_k\|_2^2 + \lambda_p \mathcal{R}_{\text{par}}(\mathbf{z}). \end{aligned}$$

The first term enforces observation consistency; the second anchors pilots; the third ties combiner outputs to the relaxed data symbols; the rest regularize partial-FFT-domain model size, combiner energy, frequency smoothness, and parity. $\Sigma_k = \sigma^2 \mathbf{I}$ recovers ordinary squared error. This objective is not intended as an identifiable physical channel estimator: the partial-FFT-domain response tensor $\mathbf{C}$ and combiner $\mathbf{W}$ are coupled through the symbols, pilots, and regularizers, but no explicit constraint forces $\mathbf{W}$ to be the inverse of $\mathbf{C}$. Its role is to show how observation-vector consistency, combiner outputs, pilots, and code constraints can be placed in one regularized problem. Multiple $(\mathbf{C}, \mathbf{W}, \mathbf{z})$ triples can explain

the same partial-FFT observations; $\mathbf{C}$ is included to expose the coupled-carrier structure, not to claim unique identification of the physical ICI matrix. Equation (9) is written for a single-symbol carrier neighborhood. Under CP-short leakage, the index can be lifted from k to (k, s) and $\mathcal{B}_K(k)$ replaced by a local space-time neighborhood that also contains neighboring OFDM symbols:

$$\mathbf{Y}_{k,s} = \sum_{(q,r) \in \mathcal{B}(k,s)} \mathbf{c}_{k,s;q,r} s_{q,r}(\mathbf{z}) + \mathbf{n}_{k,s}. \quad (10)$$

The same pilot, data-anchor, smoothness, and parity terms then regularize whether the coupling came from Doppler ICI, insufficient CP, or both. This is a modeling extension, not a CP-short experiment in this paper: it shows how code structure could add global parity constraints to an otherwise underconstrained front-end fit, but the measured results below should not be read as a validated CP-relaxation rule.

D. Relation to the Pilot-Trained Combiner

JUNA is the natural lifting of (5): the pilot-only desired response is replaced by the packet-level partial-FFT-domain model, every carrier refines the front end, the data field is generated by bit latents so soft bits improve during inference, and the code enters through (7). Equation (9) is best read as a theoretical lens—it is not the exact algorithm run on measured channels. In particular, the at-scale receivers do not estimate the full partial-FFT-domain response tensor $\mathbf{C}$ or the banded ICI matrix; those variables make explicit the physical coupling that motivates the update. The operative connection is narrower: JUNA-Wz freezes the partial-FFT-domain model and descends a reduced $(\mathbf{W}, \mathbf{z})$ objective, while JUNA-lite uses the fixed- $\mathbf{z}$ view to turn the tie term into an RLS target-fitting problem for the combiner. With the combiner fixed, BP supplies the parity-aware posterior used for the next latent-symbol or anchor update. Section V makes this alternation, and its limits, precise.

V. IMPLEMENTED RECEIVERS: JUNA-WZ AND JUNA-LITE

We implement two measured receivers derived from the coupled objective. The reduced gradient receiver, JUNA-Wz, freezes the explicit partial-FFT-domain response tensor $\mathbf{C}$ at the partial-FFT observation model and updates the partial-FFT combiner $\mathbf{W}$ together with the soft bit latents $\mathbf{z}$. Its coordinates are weighted by decoder confidence and can be projected periodically through the LDPC BP decoder (Section V-E). Because that solve repeats gradient steps and decodes per packet, we also implement JUNA-lite, a lightweight approximation that replaces explicit gradient descent with a bounded alternating posterior-anchor RLS refit (Section V-C). It keeps the same partial-FFT observation model and combiner but re-estimates the combiner *toward* decoder posterior means. JUNA-lite costs a fixed number of RLS/BP passes per packet and is cheap enough to run across every measured capture, so it carries the large-scale sweeps; JUNA-Wz is the gradient mirror used to check the approximation. Both receivers share the LDPC decode pass and the decoder-grounded candidate selection

described next, and differ only in how the combiner and data symbols move between decodes. The full $(\mathbf{W}, \mathbf{C}, \mathbf{z})$ optimizer, JUNA-WCz, is a diagnostic path described separately. Fig. 4 gives the receiver loop as a pictorial summary: the top row is the seed Partial-FFT/FEC path, and the dashed paths are the decoder-to-front-end feedback used by JUNA-lite and JUNA-Wz. Algorithms 4 and 5 list the two deployed updates in the same notation.

A. OFDM5 Prototype Solvers: Reproducible Update Forms

For clarity, we also record the optimization paths in the earlier OFDM5 prototype. These paths are useful implementation lineage, but they are not the same claim as the measured JUNA-Wz/JUNA-lite receivers above. OFDM5 contains three inner mechanisms: a block-coordinate least-squares solver for a local ICI combiner, a manually differentiated code-aware partial-FFT solver, and optional ForwardDiff-based symbol refiners. A decoder-aware wrapper around the block-coordinate solver is also summarized below, because that wrapper is the code path that supplies the pseudo-pilot vector used by the second block-coordinate pass.

Let V be the number of partial-FFT temporal views and define $\mathbf{y}_k = [Y_{1,k}, \dots, Y_{V,k}]^T$. The local ICI solver divides the FFT bins into B frequency bands,

$$b(k) = 1 + \left\lfloor \frac{(k-1)B}{N} \right\rfloor, \quad b(k) \in \{1, \dots, B\}, \quad (11)$$

and first builds the “safe” pilot-neighborhood bins

$$\Omega_K = \{k : k - \Delta \in \mathcal{P} \text{ for all } \Delta = -K, \dots, K\}, \quad (12)$$

where indexing is circular modulo N . In the ordinary pilot-only mode the training set is $\Omega_{\text{tr}} = \Omega_K$, $s_k^{\text{ref}} = s_{\text{pilot}}$, and $\rho_k = 1$. When `pseudo_enable` is true and a fixed symbol vector $\mathbf{s}^{\text{fixed}}$ is supplied, OFDM5 replaces the safe-pilot set by weighted pilot-plus-decision training,

$$\begin{aligned} \kappa_k &= \text{clip}(|s_k^{\text{fixed}}| / |s_{\text{pilot}}|, 0, 1), \\ \nu_k &= \rho_{\text{data}} \kappa_k^{p_{\text{conf}}}, \\ s_k^{\text{ref}} &= \begin{cases} s_{\text{pilot}}, & k \in \mathcal{P}, \\ s_k^{\text{fixed}}, & k \notin \mathcal{P}, \end{cases} \\ \rho_k &= \begin{cases} 1, & k \in \mathcal{P}, \\ \nu_k, & k \notin \mathcal{P}, \nu_k \geq \rho_{\min}, \\ 0, & \text{otherwise}, \end{cases} \\ \Omega_{\text{tr}} &= \{k : \rho_k > 0\}. \end{aligned} \quad (13)$$

Thus pseudo-training uses the decided symbol on a data tone itself, rather than requiring every neighboring tone to be a true pilot. For band b , the combiner is $\mathbf{w}_b \in \mathbb{C}^V$, the local ICI coefficients are $\mathbf{c}_b = [C_{-K,b}, \dots, C_{K,b}]^T$, and the pilot or pseudo-pilot reference on tone q is s_q^{ref} . With optional non-

negative training weights ρ_k , the OFDM5 weighted objective is

$$\begin{aligned}
 a_{k,\Delta} &= H_{k-\Delta} s_{k-\Delta}^{\text{ref}}, \\
 e_k &= \mathbf{w}_{b(k)}^H \mathbf{y}_k - \sum_{\Delta=-K}^K C_{\Delta,b(k)} a_{k,\Delta}, \\
 J_{\text{WC}}(\mathbf{W}, \mathbf{C}) &= \sum_{k \in \Omega_{\text{tr}}} \rho_k |e_k|^2 \\
 &\quad + \eta_W \sum_b \|\mathbf{w}_b\|_2^2 \\
 &\quad + \eta_C \sum_b \|\mathbf{c}_b\|_2^2 \\
 &\quad + \gamma_C \sum_b \|\mathbf{D}\mathbf{c}_b\|_2^2,
 \end{aligned} \tag{14}$$

where $\mathbf{D}$ is the first-difference matrix. The default OFDM5 :wc path does not differentiate this loss. It alternates two normal equation solves. Holding $\mathbf{C}$ fixed, form one row per training tone in band b ,

$$(\mathbf{Y}_b)_{k,:} = \sqrt{\rho_k} \mathbf{y}_k^T, \quad t_{b,k} = \sqrt{\rho_k} \sum_{\Delta=-K}^K C_{\Delta,b} H_{k-\Delta} s_{k-\Delta}^{\text{ref}}. \tag{15}$$

The code solves for $\mathbf{u}_b = \overline{\mathbf{w}}_b$,

$$\mathbf{u}_b = (\mathbf{Y}_b^H \mathbf{Y}_b + \eta_W \mathbf{I})^{-1} \mathbf{Y}_b^H \mathbf{t}_b, \quad \mathbf{w}_b = \overline{\mathbf{u}}_b. \tag{16}$$

Holding $\mathbf{W}$ fixed, form

$$(\mathbf{A}_b)_{k,\Delta} = \sqrt{\rho_k} H_{k-\Delta} s_{k-\Delta}^{\text{ref}}, \quad r_{b,k} = \sqrt{\rho_k} \mathbf{w}_b^H \mathbf{y}_k. \tag{17}$$

Then

$$\mathbf{c}_b = (\mathbf{A}_b^H \mathbf{A}_b + \eta_C \text{diag}(\mathbf{q}_b) + \gamma_C \mathbf{D}^T \mathbf{D})^{-1} \mathbf{A}_b^H \mathbf{r}_b, \tag{18}$$

where $\mathbf{q}_b = \mathbf{1}$ unless the sparse-support prior is active (c_sparse_enable, positive c_sparse_topk, positive c_sparse_offdiag_boost, and $2K+1 > 1$). In that case the code scores the columns by $|\mathbf{A}_b^H \mathbf{r}_b|$, always keeps the center delay, keeps the largest c_sparse_topk scores, and adds c_sparse_offdiag_boost to the other diagonal penalties. Empty bands retain the previous estimate; without a previous estimate, the $\mathbf{W}$ fallback is the uniform combiner $(1, \dots, 1)^T / \sqrt{V}$ and the $\mathbf{C}$ fallback is the center-tap identity. After each solve, OFDM5 optionally normalizes $\mathbf{w}_b \leftarrow \mathbf{w}_b / (\|\mathbf{w}_b\|_2 + \epsilon)$ and rotates/scales $\mathbf{c}_b$ by dividing by $|C_{0,b}| + \epsilon$ and multiplying by $e^{-j\angle C_{0,b}}$, so that the center coefficient is real and close to one. If a direct spectrum $\mathbf{x}^{\text{fixed}}$ is supplied, the code blends

$$\mathbf{x} = \alpha \mathbf{x}_{\text{model}} + (1 - \alpha) \mathbf{x}^{\text{fixed}}, \quad \alpha = \text{clip}(\text{xblend}, 0, 1), \tag{19}$$

and if use_w=false, $\mathbf{x}^{\text{fixed}}$ is required and the $\mathbf{W}$ -step is skipped. The demixing stage then approximately inverts the local ICI operator $\mathcal{C}$ by guarded Richardson iteration,

$$\mathbf{z}^{(r+1)} = \mathbf{z}^{(r)} + \mu (\mathbf{x} - \mathcal{C}\mathbf{z}^{(r)}), \quad \mathbf{x}_k = \mathbf{w}_{b(k)}^H \mathbf{y}_k, \tag{20}$$

with $\mu = \text{demix_safety} / \|\widehat{\mathcal{C}}\|$ chosen from a power-iteration norm estimate when μ is not supplied. The guard stores the previous iterate; if a step gives a nonfinite residual or

Algorithm 1 OFDM5 block-coordinate LS local-ICI solver

Require: Partial-FFT views $\mathbf{y}_k$, pilot mask, channel estimate H_k , optional fixed symbols $\mathbf{s}^{\text{fixed}}$, optional direct spectrum $\mathbf{x}^{\text{fixed}}$, half-bandwidth K , bands B , iterations T

- 1: Build Ω_{tr} : use (12) in pilot-only mode, or (13) in pseudo-training mode.
- 2: Initialize $\mathbf{w}_b = (1, \dots, 1)^T / \sqrt{V}$ and $C_{0,b} = 1$, $C_{\Delta \neq 0,b} = 0$.
- 3: **for** $t = 1, \dots, T$ **do**
- 4: **if** use_w and $\mathbf{W}$ is not frozen **then**
- 5: **for** each band b **do**
- 6: Build $\mathbf{Y}_b, \mathbf{t}_b$ from (15).
- 7: Solve (16); optionally normalize $\mathbf{w}_b$.
- 8: **end for**
- 9: **end if**
- 10: **if** $\mathbf{C}$ is not frozen **then**
- 11: **for** each band b **do**
- 12: Build $\mathbf{A}_b, \mathbf{r}_b$ from (17); if use_w=false, replace $r_{b,k}$ by $\sqrt{\rho_k} x_k^{\text{fixed}}$.
- 13: Solve (18); optionally normalize $\mathbf{c}_b$.
- 14: **end for**
- 15: **end if**
- 16: Record J_{WC} from (14).
- 17: **end for**
- 18: Compute $x_{\text{model},k} = \mathbf{w}_{b(k)}^H \mathbf{y}_k$, or use $\mathbf{x}^{\text{fixed}}$ when use_w=false.
- 19: If $\mathbf{x}^{\text{fixed}}$ is present, blend by (19).
- 20: **if** demixing is enabled **then**
- 21: Run (20) to obtain $\mathbf{z}$.
- 22: **else**
- 23: Set $\mathbf{z} = \mathbf{x}$.
- 24: **end if**
- 25: Optionally call the AD symbol refiner in Algorithm 3.
- 26: **return** $\mathbf{W}, \mathbf{C}, \mathbf{x}, \mathbf{z}$.

exceeds demix_guard_factor times the initial residual, the code rolls back to the previous iterate and terminates. It also stops early once the residual falls below 10^{-6} of its initial value.

The decoder-aware OFDM5 wrapper uses Algorithm 1 as an inner solver. It first runs a seed block-coordinate pass, forms data-bin LLRs from the demixed symbols after optional one-tap equalization, $\tilde{s}_i = H_i^* z_i / (|H_i|^2 + \epsilon)$, and calls BP with half-LLR messages. For QPSK pseudo-training, the BP bit probabilities are converted into soft constellation points

$$s_{d_i}^{\text{BP}} = \frac{1}{\sqrt{2}} [\text{clip}(1 - 2p_{2i-1}) + j \text{clip}(1 - 2p_{2i})], \tag{21}$$

while pilot tones are reset to s_{pilot} . The wrapper then builds $\mathbf{s}^{\text{fixed}} = \mathbf{s}^{\text{BP}}$, $\mathbf{x}^{\text{fixed}} = H_{\text{raw}} \odot \mathbf{s}^{\text{fixed}}$, and reruns the block-coordinate solver warm-started from the first pass's $\mathbf{W}, \mathbf{C}$ with pseudo-training enabled. It keeps the better decode by validity first, then syndrome weight, optionally mean $|\text{LLR}|$, and finally BP iteration count. If the optional LLR-domain refinement is enabled, OFDM5 optimizes

$$\begin{aligned}
 \eta_m(\ell) &= 1 - \prod_{r \in \mathcal{N}(m)} \text{clip}(\tanh(\ell_r/2)), \\
 J_\ell(\ell) &= \sigma^{-2} \|\mathcal{S}_\epsilon \text{diag}(H) \mathbf{X}(\ell) - \mathbf{z}\|_2^2 \\
 &\quad + \lambda \sum_m \eta_m(\ell)^2 + \gamma \|\ell\|_2^2,
 \end{aligned} \tag{22}$$

where $\mathcal{S}_\epsilon = \mathbf{F} \text{diag}(e^{j2\pi\epsilon n/N}) \mathbf{F}^H$ is applied by FFTs and $\mathbf{X}(\ell)$ is the QPSK soft-symbol map using $\tanh(\ell/2)$. This branch uses manual gradients inside an L-BFGS optimizer, then applies the same BP and decode-preference policy.

The same OFDM5 entry point also has a separate :wh mode. This branch requires use_w=true and does not esti-

mate the local ICI matrix $\mathbf{C}$. Instead it alternates the banded combiner with a length- L channel impulse response $\mathbf{h}$, with $H(\mathbf{h}) = \text{FFT}([\mathbf{h}; 0])$:

$$J_{\text{WH}}(\mathbf{W}, \mathbf{h}) = \sum_{k \in \mathcal{P}} \left| \mathbf{w}_{b(k)}^H \mathbf{y}_k - H_k(\mathbf{h}) s_{\text{pilot}} \right|^2 + \eta_W \|\mathbf{W}\|_F^2 + \mathbf{h}^H \mathbf{\Lambda}_{\text{wh}} \mathbf{h} + \mu_{\text{wh}} \|\mathbf{h}\|_1. \quad (23)$$

The delay taper used here and in SPADE is

$$[\mathbf{\Lambda}]_{\ell, \ell} = \lambda_0 \left(1 + \frac{\ell}{L_*} \right)^2, \quad (24)$$

$$L_* = \begin{cases} L_{\text{decay}}, & L_{\text{decay}} > 0, \\ \max(L/4, 1), & L_{\text{decay}} \leq 0, \end{cases}$$

for tap index $\ell = 0, \dots, L-1$; SPADE's ISTA weights use the same taper with $\lambda_0 = 1$. In `:wh` mode the `freeze_c` flag freezes this $\mathbf{h}$ -step. The implementation initializes $\mathbf{h}$ by fitting H_{raw} over all tones with a tapered ridge penalty, solves $\mathbf{W}$ on all pilot tones using targets $H_k(\mathbf{h}) s_{\text{pilot}}$, then re-estimates $\mathbf{h}$ from

$$z_k^{\text{pilot}} = (1 - \rho_{\text{anchor}}) \frac{\mathbf{w}_{b(k)}^H \mathbf{y}_k}{s_{\text{pilot}}} + \rho_{\text{anchor}} H_{\text{raw}, k}, \quad k \in \mathcal{P}. \quad (25)$$

The $\mathbf{h}$ -step is a tapered LS fit of $\mathbf{F}_{\mathcal{P}, L} \mathbf{h} \approx \mathbf{z}^{\text{pilot}}$, with optional complex soft-thresholding. The returned $\mathbf{C}$ field is only a zero placeholder in this mode; downstream code uses $H(\mathbf{h})$ and the same $\mathbf{x}^{\text{fixed}}$ blend and optional BPSK code refiner as above, but it does not run the Richardson demixing stage.

OFDM5 also contains a manually differentiated solver, named SPADE in that codebase, that fits a longer channel and QPSK soft symbols directly to the partial-FFT observation matrix. The code requires V to divide N and works with the unitary observation

$$\mathbf{Y}_u = \frac{\mathbf{Y}_p}{\text{scaleYp} \sqrt{N}}, \quad (26)$$

$$\text{scaleYp} = \begin{cases} \sqrt{V}, & V > 1, \\ 1, & V = 1. \end{cases}$$

Let $\mathcal{A}_H(\mathbf{X})$ denote the matrix-free operator that maps a frequency-domain symbol vector $\mathbf{X}$ through frequency response H , a unitary IFFT, temporal windowing into V partial-FFT views, and unitary FFTs back to the observation grid. For data tone d_i , OFDM5 parameterizes the QPSK soft symbol by logits

$$s_i^R = \text{clip}(\tanh u_{2i-1}, -c_{\text{soft}}, c_{\text{soft}}),$$

$$s_i^I = \text{clip}(\tanh u_{2i}, -c_{\text{soft}}, c_{\text{soft}}), \quad (27)$$

$$X_{d_i}(\mathbf{u}) = \frac{1}{\sqrt{2}} (s_i^R + j s_i^I).$$

while pilot tones remain fixed. For fixed H , the symbol-step

loss is

$$J_u(\mathbf{u}; H) = \sigma^{-2} \|\mathcal{A}_H(\mathbf{X}(\mathbf{u})) - \mathbf{Y}_u\|_F^2 + \lambda_{\text{anchor}} \sum_i |X_{d_i}(\mathbf{u}) - X_{\text{anchor}, d_i}|^2 + \lambda_{\text{code}} \sum_m \left(1 - \prod_{\ell \in \mathcal{N}(m)} \text{clip}(\tanh u_\ell) \right)^2 + \lambda_{\text{hard}} \sum_\ell (1 - s_\ell^2), \quad (28)$$

The implementation computes the residual $\mathbf{R} = \mathcal{A}_H(\mathbf{X}(\mathbf{u})) - \mathbf{Y}_u$, applies the adjoint partial-FFT operator to obtain

$$\mathbf{g}_X = 2\sigma^{-2} \mathcal{A}_H^*(\mathbf{R}), \quad (29)$$

and then chains through $X_{d_i}(\mathbf{u})$. For example,

$$\frac{\partial J_u}{\partial u_{2i-1}} \leftarrow \frac{1 - (s_i^R)^2}{\sqrt{2}} \Re\{g_{X, d_i}\},$$

$$\frac{\partial J_u}{\partial u_{2i}} \leftarrow \frac{1 - (s_i^I)^2}{\sqrt{2}} \Im\{g_{X, d_i}\}, \quad (30)$$

before adding the anchor, hard-symbol, and parity-product derivatives. The channel step builds a design matrix $\mathbf{G}(\mathbf{X})$ for a length- L impulse response $\mathbf{h}$ and solves a ridge problem whose $\mathbf{\Lambda}$ follows (24),

$$\mathbf{M}_h = \sigma^{-2} \mathbf{G}^H \mathbf{G} + \mathbf{\Lambda} + \lambda_{\text{ref}} \mathbf{F}_{\mathcal{P}}^H \mathbf{F}_{\mathcal{P}},$$

$$\mathbf{r}_h = \sigma^{-2} \mathbf{G}^H \text{vec}(\mathbf{Y}_u) + \lambda_{\text{ref}} \mathbf{F}_{\mathcal{P}}^H \mathbf{H}_{\text{ref}, \mathcal{P}}, \quad (31)$$

$$\mathbf{h}_{\text{ridge}} = \mathbf{M}_h^{-1} \mathbf{r}_h,$$

The λ_{ref} terms are omitted unless H_{ref} is present, $\lambda_{\text{ref}} > 0$, and the pilot set is nonempty. If ISTA is disabled, $\mathbf{h} = \mathbf{h}_{\text{ridge}}$. If ISTA is enabled, the code starts from the previous $\mathbf{h}$ when available and applies shrinkage on the tap vector:

$$\mathbf{g}_h = \mathbf{M}_h \mathbf{h} - \mathbf{r}_h, \quad (32)$$

$$h_\ell \leftarrow \mathcal{S}_{\tau \lambda_g \omega_\ell} (h_\ell - \tau [\mathbf{g}_h]_\ell).$$

Thus SPADE is a manual-gradient/proximal alternating method rather than full automatic differentiation through the receiver.

Finally, OFDM5 uses ForwardDiff only in optional bit/symbol refiners. The QPSK refiner uses the same soft-symbol map as SPADE but optimizes a local loss around the current demixed data symbols $z_{0,i}$. Let q_i be the nearest QPSK point to $z_{0,i}$ and $c_i = 1/(1 + 4|z_{0,i} - q_i|^2)$. Its ForwardDiff loss is

$$\pi_m(\mathbf{u}) = \prod_{\ell \in \mathcal{N}(m)} \tanh u_\ell,$$

$$J_{\text{AD}, \text{Q}}(\mathbf{u}) = \sum_i \lambda_{\text{anchor}} |X_i(\mathbf{u}) - z_{0,i}|^2 + \sum_i \lambda_{\text{const}} c_i |X_i(\mathbf{u}) - q_i|^2 + \lambda_{\text{code}} \sum_m (\pi_m(\mathbf{u}) - \chi_m)^2 + \gamma_z \|\mathbf{u}\|_2^2, \quad (33)$$

Algorithm 2 OFDM5 manual-gradient SPADE-style solver

Require: Partial-FFT observations $\mathbf{Y}_p$ with $V \mid N$, pilots, seed symbols $\mathbf{X}_{\text{anchor}}$, optional reference channel H_{ref} , LDPC parity checks

- 1: Form $\mathbf{Y}_u$ by (26).
- 2: Fix pilots in $\mathbf{X}_{\text{base}}$; initialize data logits $\mathbf{u}$ from $\mathbf{X}_{\text{anchor}}$.
- 3: Initialize $\mathbf{h} = \text{IFFT}(H_{\text{ref}})_{1:L}$ and $H = H_{\text{ref}}$ if available; otherwise set $\mathbf{h} = 0$ and $H = \text{FFT}(\mathbf{h})$.
- 4: Initialize Adam moments once.
- 5: **for** outer iteration $r = 1, \dots, R$ **do**
- 6: Write data symbols $\mathbf{X}(\mathbf{u})$ into $\mathbf{X}_{\text{base}}$.
- 7: Build $\mathbf{G}(\mathbf{X})$, solve (31), and optionally apply (32).
- 8: Set $H = \text{FFT}(\mathbf{h})$.
- 9: **for** symbol step $q = 1, \dots, Q$ **do**
- 10: Evaluate J_u in (28).
- 11: Compute manual gradients using (29) and (30), plus parity and anchor derivatives.
- 12: Update $\mathbf{u}$ by Adam, with moments carried across outer iterations, or by plain gradient descent if Adam is off.
- 13: **end for**
- 14: **end for**
- 15: **return** $\mathbf{X}(\mathbf{u})$, H , $\mathbf{h}$ and loss components.

where $\chi_m = 1$ for even parity-check degree and $\chi_m = -1$ for odd degree, matching the code's spin convention. The BPSK refiner is analogous:

$$\begin{aligned}
 J_{\text{AD,B}}(\mathbf{u}) = & \lambda_{\text{data}} \sum_i |x_i - H_i \tanh u_i|^2 \\
 & + \lambda_{\text{code}} \sum_m (1 - \pi_m(\mathbf{u}))^2 \\
 & + \lambda_{\text{const}} \sum_i (\tanh^2 u_i - 1)^2 + \gamma_z \|\mathbf{u}\|_2^2.
 \end{aligned} \quad (34)$$

The 8PSK refiner uses eight logits per data symbol. For constellation points $\{a_m\}_{m=1}^8$ and bit spins $\sigma_{m,r} \in \{\pm 1\}$,

$$\begin{aligned}
 p_{i,m} &= \frac{\exp u_{i,m}}{\sum_{n=1}^8 \exp u_{i,n}}, \\
 X_i(\mathbf{u}) &= \sum_{m=1}^8 p_{i,m} a_m, \\
 s_{i,r}(\mathbf{u}) &= \sum_{m=1}^8 p_{i,m} \sigma_{m,r}, \\
 J_{\text{AD,8}}(\mathbf{u}) &= \sum_i \lambda_{\text{anchor}} |X_i - z_{0,i}|^2 \\
 &+ \sum_i \lambda_{\text{const}} c_i |X_i - a_i^*|^2 \\
 &+ \lambda_{\text{code}} \sum_c \left(\prod_{\ell \in \mathcal{N}(c)} \text{clip}(s_\ell) - \chi_c \right)^2 \\
 &+ \gamma_u \|\mathbf{u}\|_2^2.
 \end{aligned} \quad (35)$$

Here a_i^* is the nearest 8PSK point and $\chi_c = -1$ for odd check degree and $+1$ for even check degree. The default 8PSK backend is the manual gradient of the same loss; the `:ad` backend calls `ForwardDiff.gradient`. In all AD cases, the implementation sets $f(\mathbf{u})$ to the selected loss, computes $\mathbf{g} = \nabla_{\mathbf{u}} f$ by `ForwardDiff.gradient(f, u)`,

Algorithm 3 OFDM5 optional symbol refiner

Require: Current data symbols, modulation, optional LDPC check sets, step count T_{AD} , Adam parameters

- 1: **if** BPSK **then**
- 2: Initialize $u_i = \text{atanh}(\text{clip}(\Re\{H_i^* x_i\}/(|H_i|^2 + \epsilon), -0.95, 0.95))$.
- 3: Set $f(\mathbf{u}) = J_{\text{AD,B}}(\mathbf{u})$ from (34).
- 4: **else if** QPSK **then**
- 5: Initialize QPSK logits from $\sqrt{2}\Re\{z_{0,i}\}$ and $\sqrt{2}\Im\{z_{0,i}\}$, clipped to $[-0.95, 0.95]$.
- 6: Set $f(\mathbf{u}) = J_{\text{AD,Q}}(\mathbf{u})$ from (33).
- 7: **else**
- 8: Initialize 8PSK logits as $u_{i,m} = -s_{\text{init}} |z_{0,i} - a_m|^2$, then subtract the block maximum.
- 9: Set $f(\mathbf{u}) = J_{\text{AD,8}}(\mathbf{u})$ from (35); use the manual softmax gradient unless `backend=:ad`.
- 10: **end if**
- 11: **for** $t = 1, \dots, T_{\text{AD}}$ **do**
- 12: Compute $\mathbf{g}_t = \text{ForwardDiff.gradient}(f, \mathbf{u}_{t-1})$, except for the default 8PSK manual backend, which uses the hand-coded gradient of the same f .
- 13: Apply the Adam update (36).
- 14: Store $f(\mathbf{u}_t)$ for diagnostics.
- 15: **end for**
- 16: **if** BPSK **then**
- 17: Overwrite data tones by $z_i = H_i \tanh u_i$.
- 18: **else**
- 19: Overwrite data tones by the QPSK or 8PSK soft constellation symbol.
- 20: **end if**
- 21: **return** refined symbols, confidence diagnostics, and loss history.

and updates the logits with Adam,

$$\begin{aligned}
 \mathbf{m}_t &= \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t, \\
 \mathbf{v}_t &= \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t \odot \mathbf{g}_t, \\
 \hat{\mathbf{m}}_t &= \mathbf{m}_t / (1 - \beta_1^t), \quad \hat{\mathbf{v}}_t = \mathbf{v}_t / (1 - \beta_2^t), \\
 \mathbf{u}_t &= \mathbf{u}_{t-1} - \alpha \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon}.
 \end{aligned} \quad (36)$$

B. LDPC Decode Pass

The same LDPC interface is used by the OFDM+FEC, Partial FFT+FEC, and JUNA receivers. Its input is a scalar equalized data grid $\hat{x}_{d,s}$, not the raw waveform. First, the receiver estimates an effective demapper scale from the outer-pilot residuals,

$$\hat{\beta}_{\text{LLR}} = \max \left\{ \frac{1}{SN_p} \sum_s \sum_{p \in \mathcal{P}} |\hat{x}_{p,s} - P_{p,s}|^2, \beta_{\text{floor}} \right\}, \quad (37)$$

with a configured floor to avoid overconfident LLRs. The quantity $\hat{\beta}_{\text{LLR}}$ is an empirical LLR scale, not a physical estimate of N_0 : it is computed from pilot residuals and includes the constellation normalization convention used by the demapper. For QPSK, the channel LLRs fed to the code are

$$\begin{aligned}
 L_{I,d,s}^{\text{ch}} &= \text{clip}_{L_{\text{max}}} \left(\frac{2\Re\{\hat{x}_{d,s}\}}{\hat{\beta}_{\text{LLR}}} \right), \\
 L_{Q,d,s}^{\text{ch}} &= \text{clip}_{L_{\text{max}}} \left(\frac{2\Im\{\hat{x}_{d,s}\}}{\hat{\beta}_{\text{LLR}}} \right),
 \end{aligned} \quad (38)$$

where positive LLR means bit zero and negative LLR means bit one. Other PSK modulations use the analogous constellation demapper. The channel LLR vector is deinterleaved into LDPC bit order, then inner-pilot message coordinates are

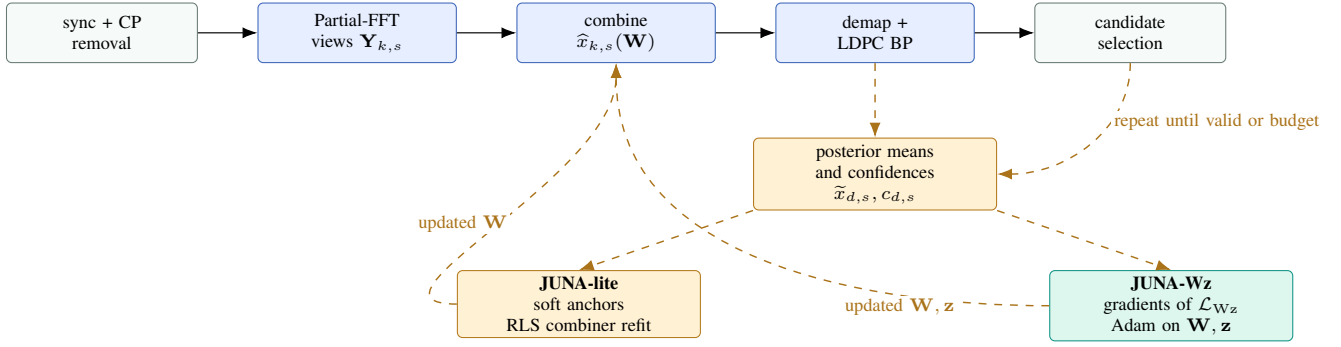


Fig. 4. Pictorial view of the implemented JUNA feedback loop. The receiver first runs the ordinary Partial-FFT/FEC seed path. LDPC BP then returns posterior soft information. JUNA-lite turns reliable posterior means into soft RLS anchors for the same Partial-FFT combiner, while JUNA-Wz uses the same decoder information inside the reduced gradient loss $\mathcal{L}_{Wz}$. Each refinement is decoded and scored, and the loop is bounded by validity checks and iteration or gradient-step budgets.

clamped to $\pm \min(L_{\max}, L_{ip})$ according to their known bits. These inner pilots are code-domain constraints; they are not OFDM pilot tones and are excluded from payload BER and PSR accounting.

The decoder then tests whether the clamped hard decision already satisfies the parity checks. If it does, the decode returns immediately with zero BP iterations and posterior LLRs equal to the clamped channel LLRs. Otherwise the code runs message passing on the Tanner graph of $\mathbf{H}_c$. In the exact sum-product path, check-to-variable messages are

$$L_{j \rightarrow i}^{(t)} = 2 \operatorname{atanh} \prod_{i' \in \mathcal{N}(j) \setminus i} \tanh \left(\frac{L_{i' \rightarrow j}^{(t-1)}}{2} \right), \quad (39)$$

and variable-to-check messages add the channel LLR and all other incoming check messages,

$$L_{i \rightarrow j}^{(t)} = L_i^{\text{ch}} + \sum_{j' \in \mathcal{M}(i) \setminus j} L_{j' \rightarrow i}^{(t)}. \quad (40)$$

The posterior returned to the receiver is $L_i^{\text{post}} = L_i^{\text{ch}} + \sum_{j \in \mathcal{M}(i)} L_{j \rightarrow i}$. The profiled workbench runs use the faster normalized min-sum approximation: each check sends the product of incoming signs times the smallest excluded magnitude, scaled by $\alpha_{\text{MS}} = 0.8$. Both variants hard-decide $b_i = 1$ when $L_i^{\text{post}} < 0$, test the syndrome after each configured check period, and stop early when all checks pass. Thus the decoder returns three objects used by JUNA: hard decoded bits for BER/validity, a syndrome-valid flag, and the posterior LLR vector that becomes soft equalizer-training information.

C. JUNA-lite: Posterior Means as Soft Data Anchors

The lightweight variant approximates the reduced code-aware update without forming a gradient. For QPSK, posterior LLRs map to posterior means and a confidence

$$\begin{aligned} \tilde{x}_{d,s} &= \frac{1}{\sqrt{2}} \left(\tanh \frac{L_I}{2} + j \tanh \frac{L_Q}{2} \right), \\ c_{d,s} &= \min \left(\left| \tanh \frac{L_I}{2} \right|, \left| \tanh \frac{L_Q}{2} \right| \right), \end{aligned} \quad (41)$$

with the obvious constellation-expectation generalization for BPSK/8PSK/16PSK. JUNA-lite forms the anchor set $\mathcal{A}_b = \mathcal{P}_b \cup \{(d, s) : d \in b, c_{d,s} \geq c_{\min}\}$ with $|\mathcal{A}_b \setminus \mathcal{P}_b| \leq A_{\max}$,

Algorithm 4 JUNA-lite posterior-anchor RLS refinement

Require: Partial-FFT views, pilots, LDPC code, budgets $I_{\max}, A_{\max}$, threshold $c_{\min}$

- 1: Train/apply the pilot-only Partial-FFT RLS seed.
- 2: Demap to channel LLRs, run LDPC BP, and store the seed candidate.
- 3: **for** $i = 1, \dots, I_{\max}$ **do**
- 4: Map posterior LLRs to $\tilde{x}_{d,s}$ and $c_{d,s}$.
- 5: **for** each RLS band b **do**
- 6: Build $\mathcal{A}_b$ from pilots and confident data anchors.
- 7: Refit the band combiner toward pilots and posterior means.
- 8: **end for**
- 9: Equalize with the updated $\mathbf{W}$ and demap to LLRs.
- 10: Decode and score the candidate by Section V-D.
- 11: Keep the best candidate under validity, syndrome, and score tests.
- 12: **if** the candidate is accepted or the budget is exhausted **then**
- 13: **break**
- 14: **end if**
- 15: **end for**
- 16: **return** the best decoder-grounded candidate.

and reruns the same RLS update with desired responses equal to the known pilots on $\mathcal{P}_b$ and the posterior means $\tilde{x}_{k,s}$ on the selected data anchors. The default receiver therefore trains toward posterior means—not hard decisions—keeps pilots in the anchor set, caps the data anchors, and can reject a refinement. Selected data anchors enter the reported RLS update with the same sample weight; the confidence value is used for gating/ranking, not as a per-sample RLS multiplier. In the reported benchmark profile $c_{\min} = 0$, so the confidence threshold itself is disabled; when the anchor cap binds, confidence is used to rank anchors, and otherwise all data tones are eligible.

D. Candidate Selection

Each pass decodes the candidate symbols and compares decoder-first. In strict mode, a valid LDPC codeword beats an invalid one, and lower syndrome weight breaks ties; remaining ties use a score combining pilot MSE, confidence-weighted distance from equalized symbols to posterior means over the data grid, normalized syndrome weight, and a small reward for sharper posteriors:

$$S_{\text{cand}} = M_{\text{pilot}} + 0.25 M_{\text{tie}} + 0.05 w_{\text{synd}} - 10^{-4} |L^{\text{post}}|. \quad (42)$$

Here

$$M_{\text{pilot}} = \frac{1}{SN_p} \sum_s \sum_{p \in \mathcal{P}} |\hat{x}_{p,s} - P_{p,s}|^2, \quad (43)$$

and, for QPSK,

$$M_{\text{tie}} = \frac{\sum_s \sum_{d \in \mathcal{D}} \max(c_{d,s}, 10^{-3}) |\hat{x}_{d,s} - \tilde{x}_{d,s}|^2}{\sum_s \sum_{d \in \mathcal{D}} \max(c_{d,s}, 10^{-3})}. \quad (44)$$

Other modulations use the same weighted tie MSE after mapping LLRs to constellation posterior means. The term w_{synd} is the syndrome weight divided by the number of parity checks, and $|L^{\text{post}}|$ is the mean posterior-LLR magnitude. Lower S_{cand} is better only after the decoder-first validity and syndrome tests above, so the scalar score does not override a valid codeword with an invalid one. The profiled replay runs enable a second, CRC-surrogate mode: the harness declares a packet accepted when the recovered unknown payload bits exactly match the transmitted payload, even if the LDPC syndrome flag is not set. This replay-only surrogate emulates an ideal payload CRC for acceptance accounting and early exit; it is not used to form posterior anchors or to change the non-oracle score in (42). In a deployed packet format, the known-payload check would be replaced by a real CRC or by strict syndrome/score selection. Strict-syndrome validity and the non-oracle score above are recorded separately.

E. JUNA-Wz: Confidence-Weighted Descent

JUNA-Wz is the deployed gradient receiver. It does not optimize the full $(\mathbf{W}, \mathbf{C}, \mathbf{z})$ objective; instead it freezes the explicit ICI/partial-FFT-domain response tensor and descends a reduced loss over the partial-FFT combiner and QPSK bit logits. Let $\hat{x}_{k,s}(\mathbf{W}) = \mathbf{w}_{b(k),s}^H \mathbf{Y}_{k,s}$ be the bandwise combiner output and $s_{k,s}(\mathbf{z})$ be the relaxed symbol implied by the bit logits. The reduced loss has the form

$$\begin{aligned} \mathcal{L}_{\text{Wz}}(\mathbf{W}, \mathbf{z}) = & \alpha_p \sum_s \sum_{p \in \mathcal{P}} |\hat{x}_{p,s}(\mathbf{W}) - P_{p,s}|^2 \\ & + \alpha_t \sum_s \sum_{d \in \mathcal{D}} c_{d,s} |\hat{x}_{d,s}(\mathbf{W}) - s_{d,s}(\mathbf{z})|^2 \\ & + \lambda_p \mathcal{R}_{\text{par}}(\mathbf{z}) + \eta_W \|\mathbf{W} - \mathbf{W}_0\|_F^2 \\ & + \gamma_z \|\mathbf{z}\|_2^2 + \frac{\mu}{2} \|\mathbf{z} - \mathbf{z}_0\|_2^2. \end{aligned} \quad (45)$$

The seed $(\mathbf{W}_0, \mathbf{z}_0)$ comes from the partial-FFT/JUNA-lite candidate and its channel or posterior LLRs, μ is a trust weight to prevent large moves away from the seed, and $\mathcal{R}_{\text{par}}(\mathbf{z})$ is (7) with $\xi_i = \tanh(z_i/2)$. The implementation evaluates analytic gradients of the combiner residuals, QPSK posterior-mean map, parity product penalty, ℓ_2 regularizer, and trust term, then applies Adam updates to $\mathbf{W}$ and $\mathbf{z}$. Each data coordinate is weighted by the decoder confidence $c_{d,s}$ of (41), so informative symbols pull harder than uncertain ones. The `wz_juna_bp` variant periodically re-decodes the refined logits and uses the BP output as a projection/checkpoint; the default `wz_juna` receiver instead selects the best decoder-grounded checkpoint across the descent. For offline diagnostics only, the harness can also record an oracle non-regression choice from known payload mismatch; this oracle choice is

Algorithm 5 JUNA-Wz robust reduced-gradient refinement

Require: Partial-FFT views, pilots, LDPC checks, seed $(\mathbf{W}_0, \mathbf{z}_0)$, confidences $c_{d,s}$, step budget G

- 1: Initialize $(\mathbf{W}, \mathbf{z}) \leftarrow (\mathbf{W}_0, \mathbf{z}_0)$.
- 2: Store the seed or JUNA-lite candidate as the current best.
- 3: **for** $g = 1, \dots, G$ **do**
- 4: Form $\hat{x}_{k,s}(\mathbf{W})$ and relaxed symbols $s_{k,s}(\mathbf{z})$.
- 5: Evaluate $\mathcal{L}_{\text{Wz}}(\mathbf{W}, \mathbf{z})$ in (45).
- 6: Compute analytic gradients with respect to $\mathbf{W}$ and $\mathbf{z}$.
- 7: Take one Adam step, including trust and regularization terms.
- 8: **if** g is a BP checkpoint **then**
- 9: Map $\mathbf{z}$ to LLRs, run BP, and score the checkpoint.
- 10: Update the best candidate by decoder-first selection.
- 11: **end if**
- 12: **end for**
- 13: Decode the final checkpoint if it was not already checked.
- 14: **return** the best decoder-grounded candidate.

not used in the receiver tables, which rely on syndrome, CRC-surrogate acceptance, or the score in (42). JUNA-lite (Section V-C) is the cheaper approximation that carries the at-scale sweeps.

F. JUNA-WCz: Full-Objective Diagnostic

JUNA-WCz updates the full $(\mathbf{W}, \mathbf{C}, \mathbf{z})$ triple of (9)—the partial-FFT combiner $\mathbf{W}$, a banded partial-FFT-domain ICI tensor $\mathbf{C}$, and the QPSK bit logits $\mathbf{z}$ —with joint Adam steps. The deployed `wz_juna` receiver fixes $\mathbf{C}$ to the front-end model and descends only $(\mathbf{W}, \mathbf{z})$. The full diagnostic loss contains the partial-FFT observation-consistency term, pilot-anchor term, combiner-symbol tie term, LDPC parity surrogate, and trust/regularization terms. Because the extra $\mathbf{C}$ degrees of freedom are numerous and only weakly identifiable from sparse pilots, this solver is not part of the large-scale receiver claim; it checks whether opening the full objective can move the same decoder metrics in the same direction on a controlled segment.

Table I reports a same-segment comparison on the current native `red_1` QPSK profile: receiver 1, replay time 0–4s, and 30 dB SNR. The comparison uses the same 11 packets and disables the usual non-regression gate, so the JUNA-WCz row is the raw optimizer output rather than “best of JUNA-lite and JUNA-WCz.” With an eight-step gradient budget and ICI half-bandwidth $K_{\text{ICI}} = 1$, the full-objective optimizer matches JUNA-lite on valid packet count (1/11 for both) and differs by only two message-bit errors out of 7491. This single segment is a diagnostic sanity check, not a replacement for the measured-channel mirrors of Section VIII; we read it as evidence that opening the full objective does not degrade decoder metrics on a controlled stress segment, not as a JUNA-WCz win over JUNA-lite.

G. Why It Helps, and Three Regimes

If posterior anchors were independent unbiased observations, enlarging the training set would reduce the variance of the RLS estimate, $\text{Cov}(\hat{\mathbf{w}}_b) \approx \sigma^2(\mathbf{Y}_{A_b}^H \mathbf{Y}_{A_b} + \delta I)^{-1}$. This is intuition, not proof: the anchors are decoder-processed functions of the same noisy seed and are correlated with it. The gain therefore comes from adding correlated but shrunken constraints that are useful when the posterior is informative;

TABLE I
RAW JUNA-LITE VERSUS DIAGNOSTIC JUNA-WCZ ON THE SAME RED_1
SEGMENT.

Method	Msg. errors	Msg. bits	Msg. BER	Code BER	Valid pkts.
JUNA-lite (posterior-anchor)	34	7491	4.539×10^{-3}	1.490×10^{-2}	1/11
JUNA-WCz (W, C, z)	32	7491	4.272×10^{-3}	1.379×10^{-2}	1/11

Receiver 1, red_1, 0–4 s, 30 dB, 11 packets, $N = 1536$, CP = 4, one symbol, pilot spacing 3, rate 1/3, $d_{\text{cfg}} = 7$, gradient steps = 8, $K_{\text{ICI}} = 1$. The hard-decision pre-BP BERs are 8.419×10^{-2} and 8.388×10^{-2} , respectively.

the danger is posterior bias, guarded by posterior means (not hard slices), anchor caps, retained pilots, validity/syndrome selection, and early stopping. This leads to three empirical regimes: on easy channels Partial FFT already decodes and JUNA ties; on impossible channels the seed posterior is uninformative and JUNA cannot help; on hard-but-decodable channels the posterior is informative but not decisive, and posterior anchors can move the re-decode across the success boundary. The extra cost is bounded for the deployed receivers: a fixed number of JUNA-Wz gradient steps or JUNA-lite RLS refits, plus decoder-grounded BP checks.

VI. COMPUTATIONAL COMPLEXITY

This section counts one decoded frame, since the replay experiments apply the same waveform schedule to all receivers. Let N be the FFT size, S the number of OFDM data symbols in a frame, N_a the number of active subcarriers, N_p the number of outer pilot tones, and $N_d = N_a - N_p$ the number of data tones. Let N_{view} be the number of partial-FFT views, B the number of RLS frequency bands, $A \leq A_{\text{max}}$ the selected posterior data anchors per symbol, and $I \leq I_{\text{max}}$ the number of JUNA refinement passes actually executed after early stopping. For JUNA-Wz, let G be the number of Adam steps in one gradient refinement and C_{ckpt} the number of BP checkpoint/projection decodes during that descent. For JUNA-WCz, let K be the ICI half-band used in the partial-FFT-domain response tensor. For the LDPC code, let $n_b = k + m$ be the codeword length, E the number of Tanner-graph edges, and T_{BP} the number of BP/min-sum iterations used by a decode attempt.

A. Common OFDM and LDPC Costs

The one-tap OFDM receiver performs packet synchronization, CP removal, an N -point FFT per OFDM symbol, pilot interpolation, symbol demapping, and one LDPC decode. The FFT term is $O(SN \log N)$. Pilot interpolation and one-tap equalization are linear in the active grid, $O(SN_a)$. The implemented LDPC decoder first checks hard parity and otherwise runs belief propagation or normalized min-sum. Both variants pass messages along Tanner edges, so the decode cost is

$$T_{\text{LDPC}} = O(T_{\text{BP}}E), \quad (46)$$

with early stopping reducing the realized iteration count. Since the systematic matrix is sparse, code construction and encoding are one-time per configuration costs outside the online receive loop; their storage and arithmetic scale with the same sparse edge count.

B. Partial-FFT RLS Front End

The Partial FFT+FEC receiver replaces each active FFT coefficient by N_{view} partial-FFT views. In the profiled implementation, each view is formed by zeroing the other short-time windows and applying the same N -point FFT plan; therefore the implemented partial-FFT extraction cost is

$$T_{\text{pfft}} = O(SN_{\text{view}}N \log N). \quad (47)$$

A shorter transform could reduce this constant, but that is not the receiver benchmarked here. Bandwise RLS training updates a length- N_{view} combiner using the outer pilots. One RLS update requires a matrix-vector multiply and rank-one inverse-covariance update, both $O(N_{\text{view}}^2)$. Training therefore costs $O(SN_p N_{\text{view}}^2)$, while applying the learned combiner to all active bins costs $O(SN_a N_{\text{view}})$. The Partial FFT+FEC receiver thus has online cost

$$T_{\text{PF}} = O(SN_{\text{view}}N \log N + SN_p N_{\text{view}}^2 + SN_a N_{\text{view}} + T_{\text{BP}}E). \quad (48)$$

For the reported runs, $N_{\text{view}} = 4$ and $B = 4$, so the RLS matrices are small; the main cost increase over one-tap OFDM is the fixed number of extra FFT views plus the LDPC decode shared by all FEC receivers.

C. Per-Pass Refinement Cost

Both deployed receivers start from the Partial FFT+FEC seed and run a bounded number of passes. JUNA-lite converts LDPC posterior LLRs into soft symbols, selects data anchors, refits the same bandwise RLS combiner using pilots plus anchors, applies the combiner, and decodes the candidate. Posterior-to-symbol conversion is linear in the data grid, $O(SN_d)$. When all QPSK data anchors are used, anchor construction remains linear. In the capped-confidence path, sorting candidates costs $O(SN_d \log N_d)$ in the worst case, but the profiled QPSK runs use the linear all-anchor path when A_{max} covers the data tones. The RLS refit costs $O(S(N_p + A)N_{\text{view}}^2)$, combiner application costs $O(SN_a N_{\text{view}})$, phase/pilot residual correction and candidate scoring are linear in the active grid, and the candidate decode costs another $O(T_{\text{BP}}E)$. Hence

$$T_{\text{JUNA}} = T_{\text{PF}} + IO(T_{\text{BP}}E + SN_d + S(N_p + A)N_{\text{view}}^2 + SN_a N_{\text{view}} + SN_a), \quad (49)$$

plus the optional $O(ISN_d \log N_d)$ sorting term when the confidence-capped path is active. This is a bounded constant-factor extension of Partial FFT+FEC under the experiment settings: $I_{\text{max}} = 4$, $A_{\text{max}} = 1024$, $N_{\text{view}} = 4$, and $B = 4$. Easy packets can terminate at the seed decode or after an early payload-exact pass; floor packets still pay at most the fixed iteration budget.

JUNA-Wz replaces the RLS refit with G Adam steps on (45). One step evaluates combiner outputs and their gradients over the active partial-FFT grid, maps logits to QPSK soft symbols, and evaluates the soft-parity gradient over Tanner edges. Its reduced-gradient increment is therefore

$$T_{\text{Wz}} = T_{\text{PF}} + O(G(SN_a N_{\text{view}} + SN_d + E) + (C_{\text{ckpt}} + 1)T_{\text{BP}}E), \quad (50)$$

TABLE II
DOMINANT ONLINE PER-FRAME RECEIVER COMPLEXITY.

Branch	Dominant cost
OFDM+FEC	$O(SN \log N + SN_a + T_{BP}E)$
Partial FFT+FEC	$O(SN_{\text{view}}N \log N + SN_p N_{\text{view}}^2 + SN_a N_{\text{view}} + T_{BP}E)$
JUNA-lite+FEC	$T_{PF} + IO(T_{BP}E + SN_d + S(N_p + A)N_{\text{view}}^2 + SN_a N_{\text{view}} + SN_a)$
JUNA-Wz+FEC	$T_{PF} + O(G(SN_a N_{\text{view}} + SN_d + E) + (C_{\text{ckpt}} + 1)T_{BP}E)$
JUNA-WCz diagnostic	$T_{PF} + O(G_{\text{full}}(SN_a(2K + 1)N_{\text{view}} + SN_d + E))$

where the extra decode covers the final decoder-grounded selection when it is not already one of the checkpoints. With fixed G , fixed checkpoint cadence, and small N_{view} , this is also a bounded online extension of Partial FFT+FEC, but with a larger constant than JUNA-lite.

JUNA-WCz is more expensive because every step additionally evaluates and updates a partial-FFT-domain response tensor with $(2K + 1)$ carrier offsets:

$$T_{WCz} = T_{PF} + O(G_{\text{full}}(SN_a(2K + 1)N_{\text{view}} + SN_d + E)). \quad (51)$$

Its memory includes $O(SBN_{\text{view}}(2K + 1))$ complex $\mathbf{C}$ coefficients in addition to $\mathbf{W}$ and the logits. This path is therefore reported as a diagnostic solver, not as the computationally inexpensive receiver used for the large measured-channel sweeps.

D. Memory and Offline Search

The one-tap receiver stores the active symbol grid, channel estimates, and LDPC messages, giving $O(SN_a + E)$ receiver memory. Partial FFT stores the partial-FFT observations, $O(SN_{\text{view}}N_a)$, plus bandwise weights and inverse covariances, $O(BN_{\text{view}} + BN_{\text{view}}^2)$. JUNA adds the current/best equalized candidates and posterior LLRs, $O(SN_a + n_b)$, but reuses the LDPC scratch buffers and RLS state across passes. Thus the additional memory over Partial FFT+FEC is linear in the frame and codeword size, not quadratic in the subcarrier count.

Configuration search is offline. If C candidate configurations are evaluated over R replay captures and Q SNR points, the brute-force search cost is $O(CRQT_{JUNA})$ for the JUNA-ranked sweeps, with smaller constants for receivers that omit refinement. The coordinate and one-at-a-time (OAT) sweeps in Section IX are therefore tuning procedures, not deployed per-packet receiver operations.

VII. EXPERIMENTAL METHOD

We evaluate three receivers on the same transmitted schedule, channel replay, SNR, FEC, and acceptance rule: **OFDM+FEC** (one-tap pilot-linear equalization), **Partial FFT+FEC** (bandwise partial-FFT RLS combining), and **JUNA-lite+FEC** (partial-FFT seed, posterior extraction, posterior-anchor RLS refinement, re-decode, candidate selection; the lightweight JUNA that carries the sweeps). Packet

success rate (PSR) is accepted packets over the scheduled replay packets entering the common decoding pipeline; the schedules are shared across receivers, and receiver-specific non-acceptance remains in the denominator. In these offline replay experiments, acceptance is payload-exact recovery, which is used as an idealized stand-in for the payload CRC available in a deployed packet format. This acceptance rule is applied uniformly to all receivers; strict LDPC validity is tracked separately but is not the headline PSR rule. BER is pooled post-FEC payload-message BER after the receiver's selection rule, excluding inner-pilot message bits. Goodput is

$$G = \frac{N_{\text{acc}}k_{\text{payload}}}{T_{\text{replay}}}, \quad (52)$$

where N_{acc} is the number of accepted packets, k_{payload} excludes inner-pilot message bits, and T_{replay} is the replay time spanned by the scheduled packets. Reported PSR uncertainty uses Wilson 95% binomial intervals computed from the accepted/decoded packet counts. Configuration search is brute force over a finite candidate pool ranked by a lexicographic tuple (PSR, goodput, accepted, payload bits, -decode time); there is no learned loss. Table IV lists the benchmark-run hyperparameters. The exact controlled and native benchmark table rows are also archived with this draft as `paper/joe/benchmark_results.csv`; the cross-channel tuning tables are generated from the reproduction pipeline in `tex/figure_data/`.

The measured data are replay-channel captures stored as red, blue, and yellow files and mapped in Section IX to SG/NA/HW sites. For each packet the transmitter generates a known preamble, outer OFDM pilot tones, data tones carrying the LDPC-coded payload, and optional inner-pilot message bits. The clean passband packet is replayed through the same measured channel segment for all receiver variants; AWGN is then added before synchronization, CP removal, demodulation, and decoding. The implemented SNR is the ratio of average clean replayed passband sample power, $N_x^{-1} \sum_n |x[n]|^2$, to added real passband noise power over the packet waveform passed to the receiver. Noise seeds are deterministic functions of the packet and SNR index unless a sweep explicitly changes them. Thus OFDM+FEC, Partial FFT+FEC, and JUNA-lite+FEC differ only in the receiver processing after the common noisy replay waveform.

Configuration identifiers encode the modem geometry and code. For example, `n1024_cp256_sym1_p3_r0p25_dc4_sig1_ip2` means $N = 1024$, CP length 256, one OFDM payload symbol per frame, one outer pilot every three active subcarriers, LDPC design rate 0.25, LDPC sparsity knob $d_{\text{cfg}} = 4$ (encoded in the suffix as `dc4`), FEC LLR scale floor 1, and one inner-pilot message bit every two message positions. For the cross-channel tuning table, the coordinate search at 20 dB sweeps FEC rate, outer/inner pilot spacing, d_{cfg} , the joint (N, CP) grid, and modulation, fixing each stage's best JUNA PSR (with BER as the tie-breaker) before moving to the next stage. The OAT table then perturbs one factor at a time around a cold default to measure coarse local sensitivity; it is used as a tuning screen, not as a causal

TABLE III
FINITE CANDIDATE POOL USED BY THE CROSS-CHANNEL COORDINATE
AND OAT TUNING SCREENS.

Knob	Candidate values
FEC rate	{0.25, 0.5, 0.75}
Outer/inner pilot ratios	coordinate: {0.10, 0.15, 0.20} ² ; OAT: {1/4, 1/5, 1/7, 1/10}
d_{cfg} (N , CP)	{6, 8, 10} {512, 1024, 2048} × {0, 16, 32, 64, 128}
Modulation	QPSK, 8PSK
Cold default	$N = 2048$, CP = 64, outer/inner = 1/4, FEC = 0.25, $d_{\text{cfg}} = 8$, QPSK

TABLE IV
RECEIVER HYPERPARAMETERS FOR THE PROFILED BENCHMARK RUNS.

Component	Value
Partial-FFT windows	4 temporal views
Partial-FFT RLS bands	4 frequency bands (unless config states otherwise)
RLS forgetting / ridge	$\lambda = 0.98$, $\delta = 10^{-2}$
FEC decoder	LDPC BP, fast min-sum in profiled runs
BP iterations	8 fast-LDPC, 24 BP cap
JUNA-lite anchor threshold / cap	$c_{\text{min}} = 0$ (off), $A_{\text{max}} = 1024$
JUNA-lite iteration budget	4 passes, early-stopped
JUNA-Wz head-to-head profile	wz_juna, Adam, 30 steps, $\mu = 50$
Acceptance	payload-exact replay acceptance (packet-CRC surrogate)

ranking when factors tie. Table III lists the pool used by these coordinate/OAT screens; the controlled benchmark uses the separate robust configuration stated in Section VIII.

Operationally, this paper calls a channel *ceiling* when Partial FFT+FEC is already above roughly 0.95 PSR, *floor* when all receivers remain near zero packet acceptance or the seed posterior is effectively uninformative, and *hard-but-decodable* when the Partial FFT seed produces useful posterior confidence but still fails payload acceptance on a substantial fraction of packets. These labels are empirical regime descriptors, not pre-deployment classifiers.

VIII. MEASURED-CHANNEL RESULTS

The SNR and stress benchmarks below run JUNA-lite, the lightweight posterior-anchor receiver, because it is cheap enough to sweep SNR, stacked receivers, and—in Section IX—twelve captures; its rows are labeled JUNA-lite+FEC. Section VIII-D and the cross-channel mirrors then report JUNA-Wz head-to-head against JUNA-lite.

A. Controlled Three-Channel Benchmark

The controlled benchmark fixes a common robust geometry (n1024_cp256_sym1_p3_r0p25_dc4_sig1_ip2) and stacked receivers {1, 2, 3} over a 0:5:20 dB sweep, on red_1 (25/5 kHz QPSK), blue_1 (13/5 kHz BPSK), and yellow_1 (13/5 kHz BPSK). Within each channel, the same waveform, FEC, and replay schedule are decoded by the three algorithms. Table V reports 20 dB. JUNA-lite+FEC gives large gains on the hard red_1 and selected blue_1 captures and a one-packet near-ceiling gain on yellow_1, with substantially lower message BER on the rescued packets—evidence that the rescued packets are genuinely better decoded, not merely re-thresholded.

TABLE V
CONTROLLED RED/BLUE/YELLOW BENCHMARK AT 20 DB.

Channel	Receiver	Payload-exact PSR	95% CI	Packets	BER	Goodput
red_1	OFDM+FEC	0.049	[0.021, 0.110]	5/102	0.193	17.8
	Partial FFT+FEC	0.559	[0.462, 0.651]	57/102	3.98×10^{-3}	202.8
	JUNA-lite+FEC	0.922	[0.853, 0.960]	94/102	6.34×10^{-4}	334.4
blue_1	OFDM+FEC	0.631	[0.538, 0.715]	70/111	0.0423	113.8
	Partial FFT+FEC	0.694	[0.603, 0.772]	77/111	6.78×10^{-3}	125.2
	JUNA-lite+FEC	0.982	[0.937, 0.995]	109/111	6.36×10^{-4}	177.3
yellow_1	OFDM+FEC	0.982	[0.936, 0.995]	108/110	2.14×10^{-4}	178.5
	Partial FFT+FEC	0.991	[0.950, 0.998]	109/110	1.07×10^{-4}	180.1
	JUNA-lite+FEC	1.000	[0.966, 1.000]	110/110	0	181.8

Goodput in bit/s. BER is pooled post-FEC payload-message BER after selection, excluding inner pilots. CI is Wilson 95% for payload-exact packet success, the replay analogue of a packet-CRC pass. Bold marks the best receiver per channel.

TABLE VI
NATIVE-MAXIMUM-BANDWIDTH FIVE-COLOR BENCHMARK AT 20 DB.

Channel	Receiver	Payload-exact PSR	95% CI	Packets	BER	Goodput
red_1	OFDM+FEC	0.000	[0.000, 0.020]	0/190	0.274	0
	Partial FFT+FEC	0.095	[0.061, 0.145]	18/190	1.69×10^{-3}	216.7
	JUNA-lite+FEC	0.711	[0.642, 0.770]	135/190	2.75×10^{-4}	1625.2
blue_1	OFDM+FEC	0.091	[0.050, 0.159]	10/110	0.178	97.9
	Partial FFT+FEC	0.318	[0.239, 0.410]	35/110	2.04×10^{-3}	342.5
	JUNA-lite+FEC	0.818	[0.736, 0.879]	90/110	3.55×10^{-4}	880.8
yellow_1	OFDM+FEC	0.893	[0.831, 0.934]	125/140	2.48×10^{-5}	1397.7
	Partial FFT+FEC	0.929	[0.874, 0.961]	130/140	9.93×10^{-5}	1453.7
	JUNA-lite+FEC	1.000	[0.973, 1.000]	140/140	0	1565.5
purple_1	all three	1.000	[0.975, 1.000]	150/150	0	1616.5
green	all three	0.000	[0.000, 0.037]	0/100	≈ 0.50	0

Goodput in bit/s. BER is pooled post-FEC payload-message BER after selection, excluding inner pilots. CI is Wilson 95% for payload-exact packet success, the replay analogue of a packet-CRC pass. purple_1/green rows collapsed (identical across receivers).

B. Native-Bandwidth Five-Color Stress Test

The second benchmark runs each color at its native high-rate profile over 0:5:30 dB—a stress test closer to the goodput frontier. Table VI reports 20 dB. The gain pattern is the predicted one: large on hard-but-decodable red_1 (18/190 → 135/190) and the selected blue_1 (35/110 → 90/110), near-ceiling on yellow_1 and purple_1, and absent on the floor channel green, where all three receivers fail.

C. JUNA-lite Mechanism Checks

Three single-run ablations of the JUNA-lite receiver were used as sanity checks rather than confidence-interval evidence. (i) Replacing posterior-mean anchors with hard-decision anchors does not destroy the gain, so the essential mechanism is decoder-derived anchoring plus decoder-grounded selection; posterior means remain the default because they avoid committing to a hard slice. (ii) Forcing the final refinement instead of selecting the best candidate is essentially tied, so selection is not the sole source of the gain. (iii) Giving the Partial-FFT seed the same LDPC decode-call budget as JUNA-lite, without posterior anchors, is numerically identical to ordinary Partial FFT in the deterministic replay path; this mainly rules out “more decode calls” as the mechanism. Multi-seed numeric ablation tables remain future work.

D. JUNA-Wz: Head-to-Head Mirror

JUNA-Wz of Section V-E is the reduced gradient receiver that JUNA-lite approximates. We therefore mirror the headline benchmarks with JUNA-Wz head-to-head against JUNA-lite

TABLE VII

JUNA-Wz VERSUS JUNA-LITE: PAYLOAD-EXACT PSR ACROSS SNR ON THE THREE REPRESENTATIVE CAPTURES. **BOLD** MARKS A GRADIENT WIN WITH DISJOINT WILSON 95% INTERVALS (GENERATED FROM `TEX/FIGURE_DATA/TAB_GRAD_VS_LITE.TEX`).

SNR	SG-1		NA-1		HW-1	
	lite	grad	lite	grad	lite	grad
0 dB	0.09	0.86	0.09	0.45	0.66	0.92
5 dB	0.71	0.99	0.53	0.82	0.91	0.98
10 dB	0.90	1.00	0.80	0.92	0.94	0.98
15 dB	0.95	1.00	0.82	0.91	0.98	1.00
20 dB	0.95	1.00	0.87	0.95	0.98	1.00
25 dB	0.94	1.00	0.88	0.95	0.96	1.00
30 dB	0.94	1.00	0.88	0.95	0.97	1.00

TABLE VIII

JUNA-Wz VERSUS JUNA-LITE: STACKED PAYLOAD-EXACT PSR SUMMARY OVER ALL 48 CAPTURE-SNR CELLS IN THE HEAD-TO-HEAD COMPARISON (GENERATED FROM `TEX/FIGURE_DATA/TAB_GRADAggregate.TEX`).

Quantity	Value
Stacked cells	48
JUNA-Wz $\geq$ JUNA-lite	44/48
CI-disjoint wins / losses	18/0
Non-disjoint regressions	4
Mean Δ PSR	+0.103

under the same payload-exact PSR rule. Table VII sweeps SNR on the three representative captures: JUNA-Wz matches or exceeds JUNA-lite at every displayed point, with the largest margins at low SNR where the seed posterior is least decisive (SG-1 rises from 0.09 to 0.86 at 0 dB) and a climb to payload-exact 1.0 as SNR grows. These displayed cases are the ceiling-reaching SG-1/NA-1/HW-1 captures; the four small non-significant regressions quantified below occur on other near-floor NA captures. Table VIII reports the full stacked summary: across all 48 capture $\times$ SNR cells, JUNA-Wz is at least as good as JUNA-lite in 44, strictly better with disjoint Wilson 95% intervals in 18, and never significantly worse.

The gain is not an artefact of the two gradient guardrails. Table IX shows that replacing the soft confidence weighting with hard anchoring, or taking the final iterate instead of the decoder-grounded checkpoint, leaves payload-exact PSR unchanged to within a single packet on the representative captures—so the improvement comes from the decoder-aided descent itself, not from confidence weighting or selection alone. The cross-channel study of Section IX extends the tuning mirror to all twelve captures; it supports the same tuning conclusions but does not show the same PSR ceiling on every capture.

TABLE IX

JUNA-Wz ABLATION AGAINST JUNA-LITE: PAYLOAD-EXACT PSR FOR THE DEFAULT SOFT Wz_JUNA, A HARD-ANCHOR VARIANT, AND A FORCED-FINAL VARIANT, EACH AGAINST JUNA-LITE, ON REPRESENTATIVE SG-1/NA-1/HW-1 CAPTURES AT 5/10 dB. PSR FOLLOWS THE SAME PAYLOAD-EXACT, POST-SELECTION DEFINITION AS THE MAIN TABLES; THE FINAL COLUMN GIVES THE PACKET DENOMINATOR.

Channel	SNR	JUNA-lite	Soft (wz)	Hard-anchor	Forced-final	Pkts.
SG-1	5 dB	0.714	0.993	0.993	0.993	140
SG-1	10 dB	0.900	1.000	1.000	1.000	140
NA-1	5 dB	0.527	0.818	0.818	0.827	110
NA-1	10 dB	0.800	0.918	0.918	0.918	110
HW-1	5 dB	0.911	0.976	0.976	0.976	124
HW-1	10 dB	0.935	0.984	0.984	0.984	124

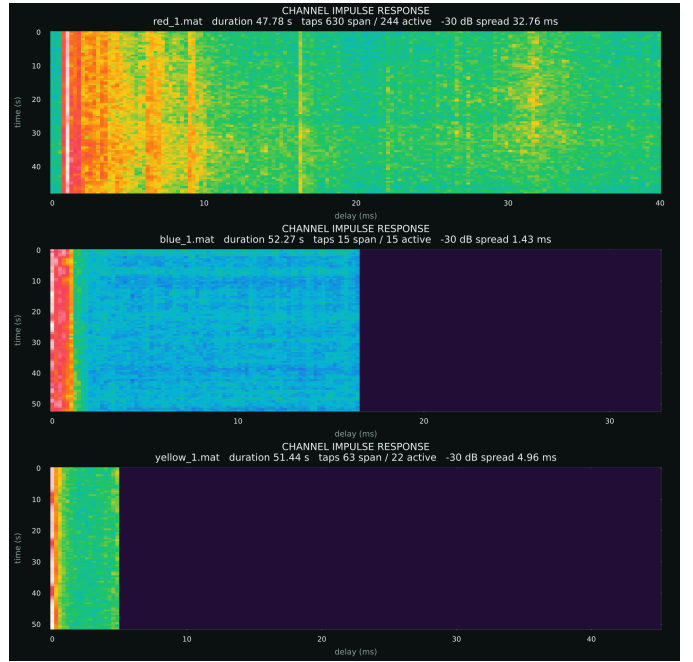


Fig. 5. Representative measured time-varying CIR per site (top to bottom: SG, NA, HW). Horizontal position is replay time, vertical position is excess delay, and color is normalized tap magnitude.

IX. CROSS-CHANNEL TUNING STUDY

The benchmarks above fix a geometry. A practical question is which modem knobs must be retuned first on a new measured channel. We replay twelve captures from three physically distinct sites—Singapore harbour (SG-1/2/3, shallow, 25 kHz), North Atlantic (NA-1/2/3, mobile deep-water, 13 kHz), and Hawaii (HW-1...6, moored deep-water, 13 kHz)—and, for each, run a coordinate search at 20 dB and one-factor-at-a-time (OAT) sensitivity sweeps at 10/20/30 dB. Fig. 5 shows a representative measured CIR per site. This is a tuning-screen study, not a train-on-one/test-on-another transfer protocol.

Two findings emerge (per-channel tables in Tables X and XI, generated by the reproduction pipeline). Table X should be read as a coarse coordinate-search screen: the packet denominators are shown explicitly, and saturated 1.000 entries over short screens are evidence of a viable local configuration, not confidence interval claims. First, QPSK is the stable

modulation across all twelve best configurations, while FEC rate 0.5 appears on the SG/NA captures and the lower-rate Hawaii configurations use denser pilots and FEC rate 0.25. The best table therefore supports conservative modulation and site-dependent pilot/FEC tuning, not a universal outer-pilot default. Second, the knobs do not all matter at once: the parameter that most often must be retuned *first* is the FFT size N —from the cold default, changing N moves packet success the most on the harder SG and NA captures. Once N is chosen well, FEC and modulation become local failure knobs; the OAT table reports mean PSR ranges from coarse one-factor screens, so equal or nearly equal FEC and modulation swings should be read as saturation or local sensitivity ties rather than evidence that the two axes are causally identical. The practical guidance is therefore two-stage: search N until packets begin decoding, then keep FEC and modulation conservative enough that the receiver stays there.

A. JUNA-Wz: Cross-Channel Tuning Mirror

Repeating the coordinate search and OAT screens with JUNA-Wz provides a gradient-side tuning mirror, not a train-on-one/test-on-another transfer experiment. Table XII mirrors Table X: QPSK is again the stable modulation, the stronger HW configurations again use denser pilots and conservative FEC, and JUNA-Wz reaches payload-exact ceiling on SG-1, NA-1, HW-1, and HW-4. The harder captures do not all saturate—for example SG-2/3, NA-2/3, and HW-5/6 remain below PSR=1—so the correct conclusion is shared tuning structure, not universal gradient ceiling. Table XIII mirrors the determinant ranking of Table XI, with FFT size N again the first-order knob on the harder SG and NA captures.

TABLE X

BEST COORDINATE-SEARCH CONFIGURATION FOUND PER REPLAY CAPTURE AT 20 DB. PSR IS PAYLOAD-EXACT REPLAY ACCEPTANCE, BER IS POOLED POST-FEC PAYLOAD-MESSAGE BER AFTER SELECTION, AND *acc./pkts.* GIVES THE ACCEPTED/SCHEDULED PACKET DENOMINATOR. THE PILOT COLUMN IS THE OUTER FREQUENCY-DOMAIN PILOT FRACTION (GENERATED FROM `TEX/FIGURE_DATA/TAB_BEST.TEX`).

channel	N	CP	outer pilot frac.	FEC	d_{cfg}	mod	payload-exact PSR	BER	acc./pkts.	rate (bit/s)
SG-1	2048	64	0.10	0.5	8	QPSK	1.000	0	10/10	5540
SG-2	512	16	0.20	0.5	8	QPSK	1.000	0	10/10	3078
SG-3	2048	64	0.20	0.5	8	QPSK	1.000	0	1/1	1308
NA-1	512	0	0.20	0.5	6	QPSK	1.000	0	10/10	2751
NA-2	512	0	0.14	0.5	8	QPSK	1.000	0	1/1	793
NA-3	512	0	0.14	0.5	8	QPSK	1.000	0	1/1	428
HW-1	2048	64	0.25	0.25	8	QPSK	1.000	0	10/10	1565
HW-2	2048	64	0.25	0.25	8	QPSK	1.000	0	10/10	3985
HW-3	2048	64	0.25	0.25	8	QPSK	1.000	0	10/10	3303
HW-4	2048	64	0.25	0.25	8	QPSK	1.000	0	10/10	1546
HW-5	512	0	0.20	0.5	6	QPSK	1.000	0	10/10	2254
HW-6	1024	0	0.20	0.5	6	QPSK	1.000	0	10/10	2024

TABLE XI

MEAN PAYLOAD-EXACT PSR SWING FROM COARSE ONE-FACTOR-AT-A-TIME SENSITIVITY SCREENS. BOLD ENTRIES ARE THE LARGEST LOCAL SWING PER CHANNEL; THE “OUTER PILOT” ROW VARIES THE FREQUENCY-DOMAIN PILOT FRACTION (GENERATED FROM `TEX/FIGURE_DATA/TAB_OAT.TEX`).

	SG-1	SG-2	SG-3	NA-1	NA-2	NA-3	HW-1	HW-2	HW-3	HW-4	HW-5	HW-6
N	.77	.79	.61	.54	.64	.27	.00	.04	.32	.00	.32	.68
d_{cfg}	.60	.19	.39	.21	.00	.00	.12	.00	.62	.09	.19	.05
outer pilot	.49	.11	.09	.24	.00	.00	.02	.02	.12	.00	.09	.12
FEC	.23	.02	.22	.46	.00	.00	.31	.17	.60	.33	.26	.00
mod	.23	.02	.22	.46	.00	.00	.45	.52	.62	1.0	.26	.07
CP	.16	.02	.08	.18	.00	.00	.02	.08	.31	.02	.18	.07

TABLE XII

JUNA-WZ MIRROR OF TABLE X: BEST COORDINATE-SEARCH CONFIGURATION PER REPLAY CAPTURE AT 20 DB FOR THE WZ_JUNA RECEIVER. PSR IS PAYLOAD-EXACT REPLAY ACCEPTANCE AND *acc./pkts.* GIVES THE ACCEPTED/SCHEDULED PACKET DENOMINATOR (GENERATED FROM `TEX/FIGURE_DATA/TAB_BEST_GRAD.TEX`).

channel	N	CP	outer pilot frac.	FEC	d_{cfg}	mod	payload-exact PSR	BER	acc./pkts.	rate (bit/s)
SG-1	1024	32	0.20	0.5	10	QPSK	1.000	0	360/360	5540
SG-2	1024	64	0.20	0.5	10	QPSK	0.600	6.0×10^{-2}	210/350	3232
SG-3	1024	64	0.20	0.5	10	QPSK	0.570	3.6×10^{-2}	85/149	1308
NA-1	1024	0	0.20	0.5	8	QPSK	1.000	0	220/220	2751
NA-2	512	0	0.20	0.5	6	QPSK	0.681	1.2×10^{-2}	113/166	793
NA-3	512	32	0.20	0.5	10	QPSK	0.374	8.4×10^{-2}	61/163	428
HW-1	2048	64	0.25	0.25	8	QPSK	1.000	0	140/140	1565
HW-2	1024	16	0.14	0.5	8	QPSK	0.963	7.9×10^{-3}	260/270	3985
HW-3	1024	32	0.14	0.5	8	QPSK	0.962	7.1×10^{-3}	250/260	3848
HW-4	2048	64	0.25	0.25	8	QPSK	1.000	0	140/140	1546
HW-5	512	16	0.20	0.5	10	QPSK	0.688	1.1×10^{-1}	330/480	2253
HW-6	512	16	0.14	0.5	8	QPSK	0.583	9.3×10^{-2}	280/480	2026

TABLE XIII

JUNA-WZ MIRROR OF TABLE XI: MEAN PAYLOAD-EXACT PSR SWING FROM ONE-FACTOR-AT-A-TIME SCREENS FOR THE WZ_JUNA RECEIVER. BOLD ENTRIES ARE THE LARGEST LOCAL SWING PER CHANNEL (GENERATED FROM `TEX/FIGURE_DATA/TAB_OAT_GRAD.TEX`).

	SG-1	SG-2	SG-3	NA-1	NA-2	NA-3	HW-1	HW-2	HW-3	HW-4	HW-5	HW-6
N	.74	.88	.61	.55	.65	.28	.00	.03	.22	.00	.37	.68
d_{cfg}	.58	.26	.46	.27	.00	.00	.07	.00	.64	.07	.29	.07
outer pilot	.53	.16	.10	.27	.00	.00	.00	.00	.07	.00	.14	.14
FEC	.26	.00	.25	.46	.00	.00	.29	.21	.71	.21	.21	.00
mod	.26	.00	.25	.46	.00	.00	.43	.57	.71	1.0	.21	.07
CP	.16	.05	.12	.27	.00	.00	.07	.08	.21	.00	.20	.07

X. LIMITATIONS

The evidence supports the receiver-level claim that JUNA-lite+FEC beats Partial FFT+FEC on measured hard-but-decodable replay channels under the tested profiles, with the expected ties and failures on ceiling and floor channels. JUNA-Wz provides supportive head-to-head mirrors but also four small, non-significant regressions on near-floor NA cells. The mechanism checks are single-run sanity ablations, not confidence-interval studies over noise seeds; the broader blue family remains capture- and modulation-sensitive and should not be presented as universally solved; and the reported BER is receiver-output BER after the selection rule, not a pure equalizer-quality measurement. The replay acceptance rule uses known-payload matching as the offline analogue of a payload CRC; strict LDPC-syndrome validity is tracked separately in the implementation. We also do not yet compute a per-capture off-diagonal-energy or banded-ICI metric, so dynamic non-diagonality is used here as a physical diagnostic rather than as a quantified predictor of JUNA gain. The soft-pilot interpretation is tested indirectly through receiver comparisons, not through a dedicated pilot-density ablation, and the CP-short space-time model is not validated as a CP-length design rule. The JUNA-WCz solver is exercised on a single segment here; the multi-capture head-to-head uses the deployed JUNA-Wz solver, and a same-profile multi-seed optimizer benchmark over the full $(\mathbf{W}, \mathbf{C}, \mathbf{z})$ objective remains future work. Separating confidence thresholding, anchor caps, early stopping, and CRC-surrogate acceptance over multiple seeds remains future work.

XI. CONCLUSION

JUNA is a practical joint receiver-decoder family for UWA OFDM under residual Doppler. The motivation is the loss of diagonal OFDM structure and the partial-FFT representation of local time variation; the coupled objective makes the code-channel feedback principle explicit, JUNA-Wz realizes it as a reduced confidence-weighted gradient descent over $(\mathbf{W}, \mathbf{z})$, and JUNA-lite recovers most of the value without forming a gradient. JUNA-lite reuses partial-FFT RLS, extracts LDPC posterior means, feeds soft anchors back into the equalizer, and accepts only refinements that improve decoder-grounded criteria, which is what makes the twelve-capture study tractable. On measured replay channels this is enough to beat Partial FFT+FEC substantially in the hard-but-decodable regime while preserving ties and failures on ceiling and floor channels, and the twelve-capture tuning study yields a simple two-stage tuning rule that both receivers broadly support. The scientific point is that decoder information is not merely a post-processing cleanup stage: in the right regime, it is equalizer training information.

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